آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Notes from E-Marking Centre on HSSC-II Mathematics Annual Examinations 2024

Introduction

This document has been produced for the teachers and candidates of Higher Secondary School Certificate (HSSC) Part II Mathematics. It contains comments on candidates' responses to the 2024 HSSC-II Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses that support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfill the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations

Most candidates achieved success in constructing good responses related to the concept of Composite functions, Differentiation (functions involving parametric equations), Integration (functions involving Partial fractions), Analytical Geometry, Conic Sections and Vectors (magnitude).
However, it is essential for teachers to concentrate on the following content and provide candidates with more practice to foster a strong understanding;

- Limit of the functions
- Differentiation (functions involving logarithms)
- Integration (Trigonometric Substitutions and Differential Equations)
- Application of higher-order calculus (Taylor series expansion, advanced differentiation)
- Linear Programming
- Rotation and transformation of axes (hyperbola transformations)
- Vector calculations (angle between vectors, dot product application)

Note: Candidates' responses shown in this report have not been corrected for grammar, spelling, format, or information.

DETAILED COMMENTS
Constructed Response Questions (CRQs)

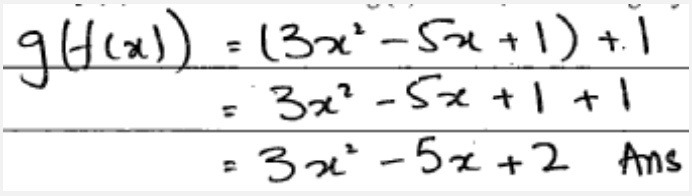
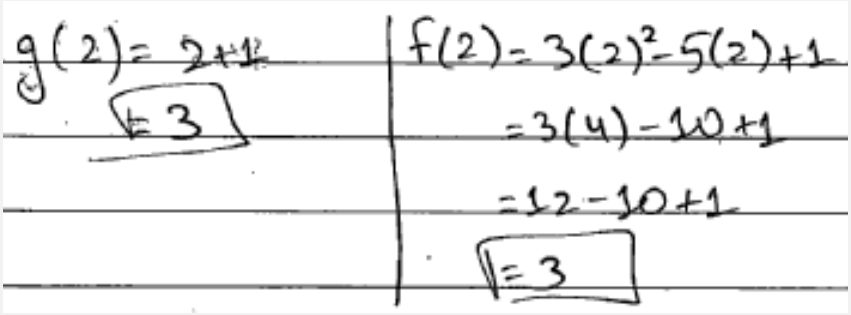
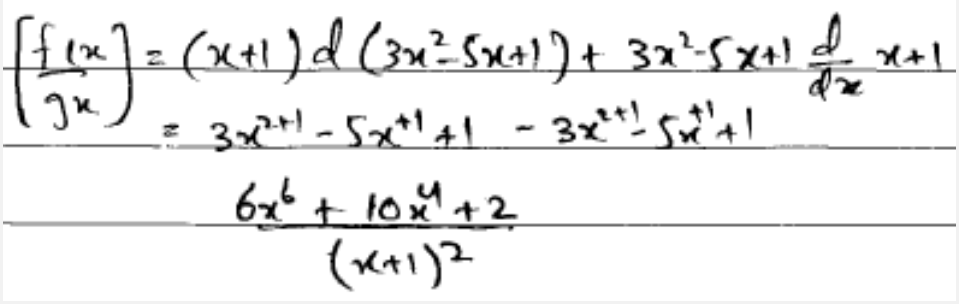
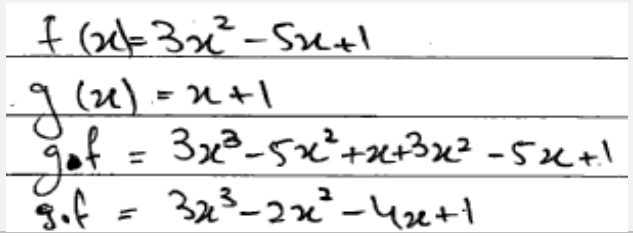
Question No. 1i	
Question Text	If $f(x) = 3x^2 - 5x + 1$ and $g(x) = x + 1$, then find $g \bullet f$.
SLO No.	12.4.2
SLO Text	Find the composition of two functions.
Max Marks	1
Cognitive Level	A*
Checking Hints	1 mark for substituting $f(x) = 3x^2 - 5x + 1$ for x in the function $g(x)$
Overall Performance	The overall performance of the candidates on this question shows that many of the candidates demonstrated a strong understanding of the underlying concept and few of the candidates struggled to apply their knowledge effectively. The results indicate a need for targeted instructional strategies to bridge the gap between understanding and application.
Description of Better Responses	Candidates who performed well on this question exhibited clarity and precision in their answers, effectively addressing all parts of the question. These candidates demonstrated a strong grasp of the concept of function composition and applied the rules for finding $g(f(x))$ in a detailed, step-by-step manner. They correctly substituted $f(x)$ into $g(x)$ and simplified the resulting expression accurately.
Image of Better Response	 $g(f(x)) = (3x^2 - 5x + 1) + 1$ $= 3x^2 - 5x + 1 + 1$ $= 3x^2 - 5x + 2$ Ans
Description of Weaker Responses	Candidates who struggled with this question typically displayed a lack of clarity, not fully addressing the requirements of the question. Their responses show inappropriate substitutions, where candidates either substituted incorrectly or used incorrect functions. Simplification errors were common, as were misidentifications of the function composition, with some of the candidates calculating $f(g(x))$ instead of $g(f(x))$.
Images of Weaker Responses	**Image (i)**  $g(2) = 2+1$ $= 3$ | $f(2) = 3(2)^2 - 5(2) + 1$ $= 3(4) - 10 + 1$ $= 12 - 10 + 1$ $= 3$ **Image (ii)**  $\left[\frac{f(x)}{g(x)}\right] = (x+1)\frac{d}{dx}(3x^2 - 5x + 1) + 3x^2 - 5x + 1 \frac{d}{dx}x + 1$ $= 3x^{2+1} - 5x^{1+1} + 1 - 3x^{2+1} - 5x^{1+1} + 1$ $\frac{6x^6 + 10x^4 + 2}{(x+1)^2}$

	Image (iii) 

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy** Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration ** For description of each Pedagogy, refer to Annexure A	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:
By implementing these strategies, educators can enhance students' understanding and application of complex concepts, ultimately improving overall performance in assessments;

Practice Using Interactive Digital Tools:

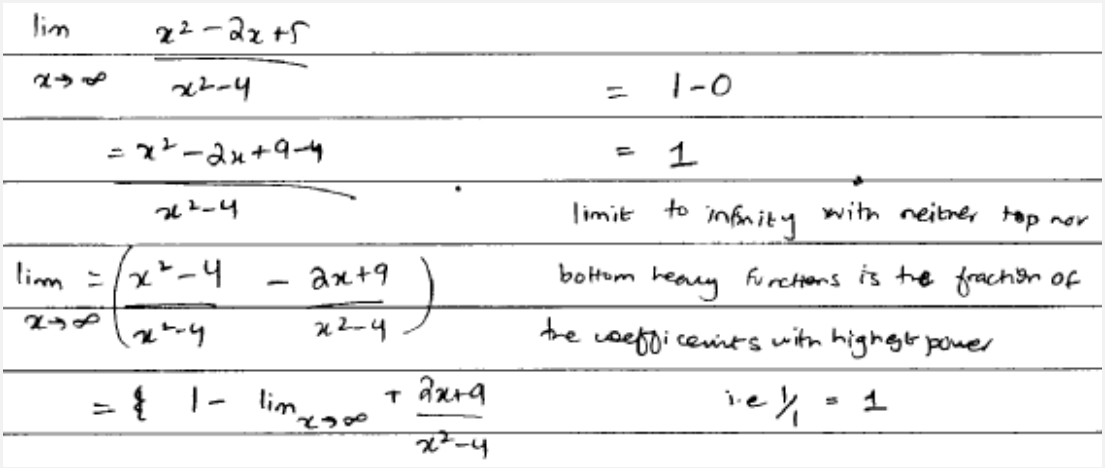
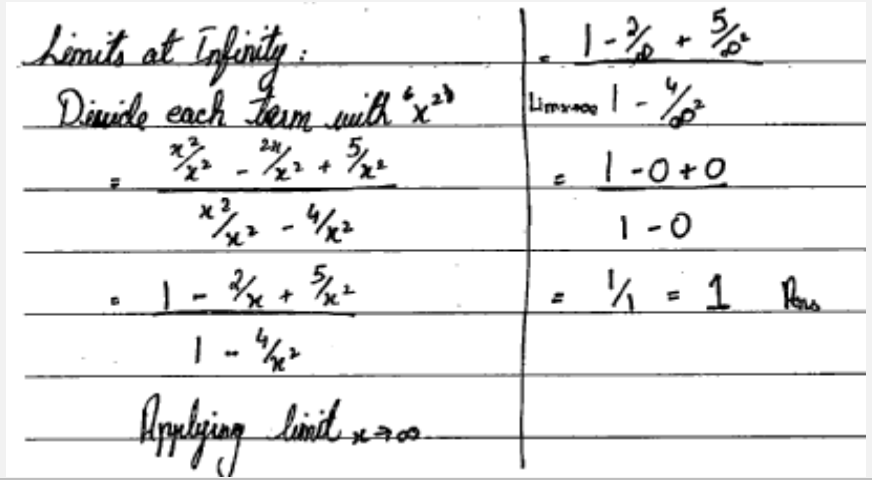
- **Desmos:** An interactive graphing calculator that allows students to visualise functions and their compositions dynamically.
- **GeoGebra:** A dynamic mathematics software that supports geometry, algebra and calculus, enabling students to explore and visualise mathematical concepts.

Discuss Inter-Disciplinary Practical Examples:

- **Physics:** In physics, composite functions can describe the relationship between different physical quantities. For example, the motion of an object can be described by composing velocity and acceleration functions.
- **Computer Science:** In programming, function composition is used in functional programming languages to build complex operations from simpler functions. This helps in modularising code and making it more readable and maintainable.

*K = Knowledge U = Understanding A = Application and other higher-order cognitive skills

Question No. 1ii	
Question Text	Evaluate $\lim_{x\to\infty} \frac{x^2-2x+5}{x^2-4}$ when $x \neq \pm 4$.
SLO No.	12.9.2
SLO Text	Evaluate limits of different algebraic, exponential, logarithmic and trigonometric functions.
Max Marks	2

Cognitive Level	A
Checking Hints	1 mark for taking x^2 common and division 1 mark for substituting ∞
Overall Performance	The overall performance on this question shows that notable candidates have demonstrated a strong understanding of evaluating limits at infinity while, many candidates struggled with the concept. Some of the candidates who performed well were able to compute the limit correctly and use algebraic simplifications effectively. Conversely, weaker responses indicate a lack of familiarity with the concept of limits at infinity, leading to incorrect methods and calculations.
Description of Better Responses	Better response answered this question correctly demonstrated a clear understanding of evaluating limits at infinity. These responses include accurate calculations and appropriate use of algebraic simplifications. For example, candidates correctly identified that the highest degree terms in the numerator and denominator dominate as x approaches infinity, simplifying the fraction to x^2/x^2 and concluding that the limit is 1. This approach reflected a thorough grasp of the concept and mathematical operations.
Images of Better Responses	**Image (i)**  $\lim_{x\to\infty} \frac{x^2-2x+5}{x^2-4} = \frac{x^2-2x+9-4}{x^2-4}$ $\lim_{x\to\infty} = \left(\frac{x^2-4}{x^2-4} - \frac{2x+9}{x^2-4}\right)$ $= \{1 - \lim_{x\to\infty} \frac{2x+9}{x^2-4}$ $= 1-0$ $= 1$ limit to infinity with neither top nor bottom heavy functions is the fraction of the coefficients with highest power i.e $1/1 = 1$ **Image (ii)**  Limits at Infinity: Divide each term with x^2 $= \frac{x^2/x^2 - 2x/x^2 + 5/x^2}{x^2/x^2 - 4/x^2}$ $= \frac{1 - 2/x + 5/x^2}{1 - 4/x^2}$ Applying limit $x\to\infty$ $= 1 - 2/\infty + 5/\infty^2$ $\lim_{x\to\infty} 1 - 4/\infty^2$ $= 1 - 0 + 0$ $1 - 0$ $= 1/1 = 1$ Ans
Description of Weaker Responses	Candidates with weaker responses often displayed several key issues. Many could not recall or apply the concept of evaluating limits at infinity correctly. Some of the candidates engaged in unnecessary factorisation or replaced x with a finite number, which led to incorrect results. Additionally, a few candidates differentiated the numerator and denominator and then applied the limit, an approach that is not valid for this type of problem. These errors indicate students' misunderstanding related to the proper methods for evaluating limits at infinity.

Images of Weaker Responses

Image (i)

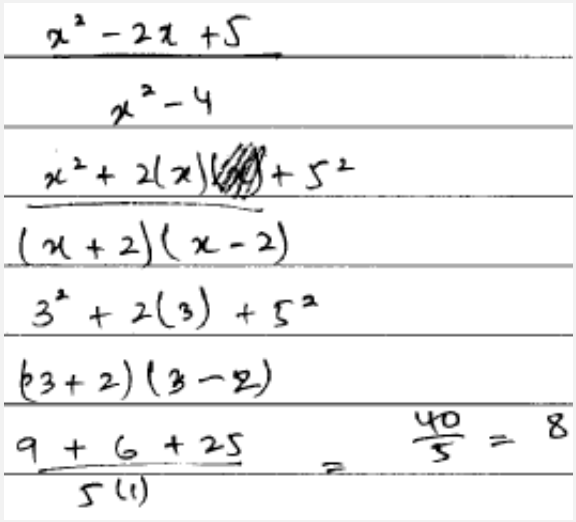


Image (ii)

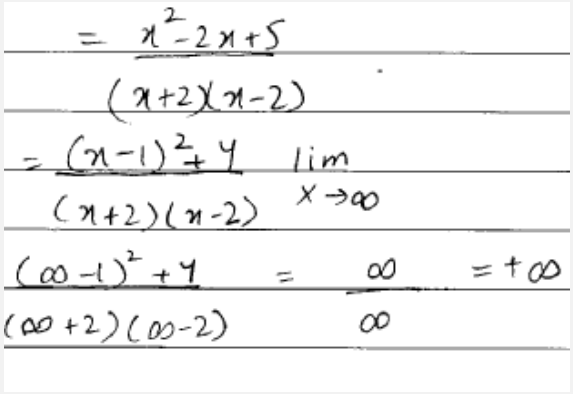


Image (iii)

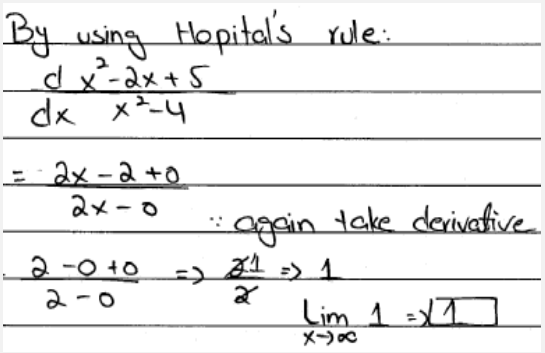
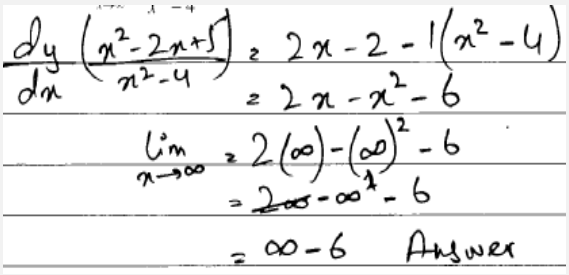


Image (iv)



Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

By implementing these strategies, educators can enhance students' understanding of limits and improve their performance;

Practice Using Interactive Digital Tools:

- **Desmos:** Use Desmos to graph the function and observe its behavior as *x* approaches infinity. This can help students visualise the limit and understand the effect of highest degree terms on the result.

- **Wolfram Alpha:** This computational tool can be used to verify limits and perform algebraic manipulations, providing students with immediate feedback on their solutions.

Real-Life Practical Examples:

- **Engineering:** In engineering, limits at infinity can be used to analyse the behavior of systems as parameters grow very large, such as in stability analysis of control systems.
- **Economics:** Limits at infinity can model scenarios where economic quantities grow without bound, such as predicting long-term behavior of growth models.
- **Population Studies:** Limits at infinity can help in understanding population growth models where growth rates become dominant over time, aiding in long-term predictions.

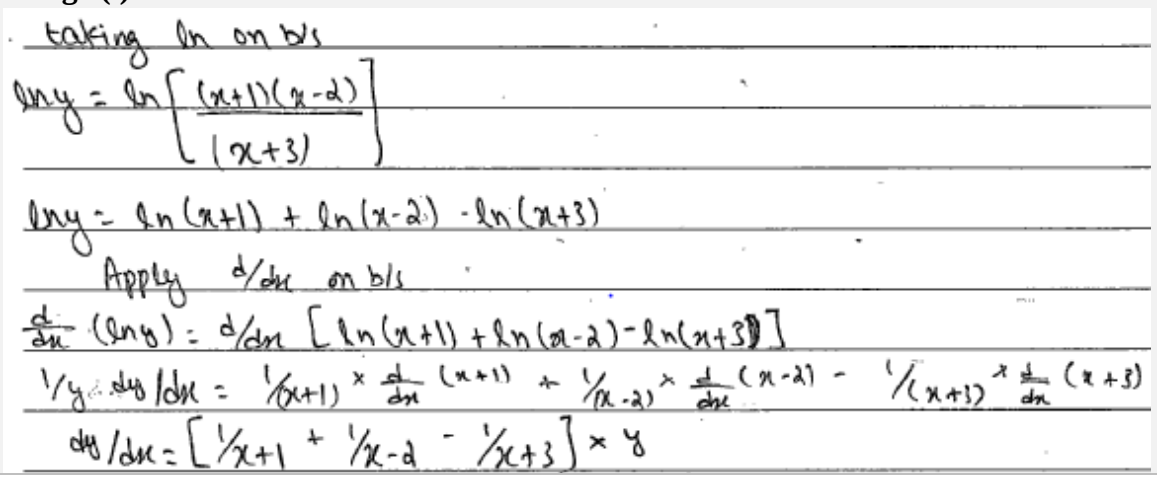
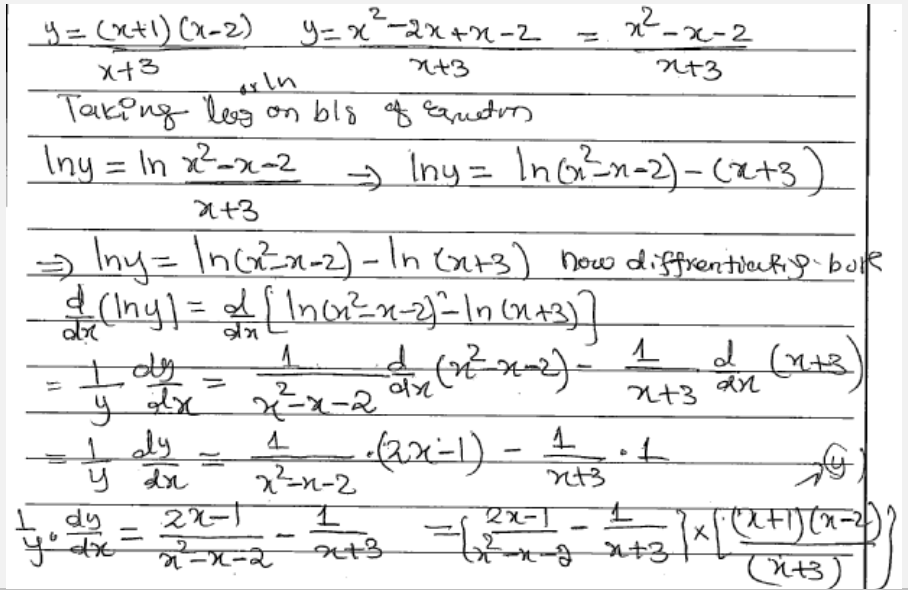
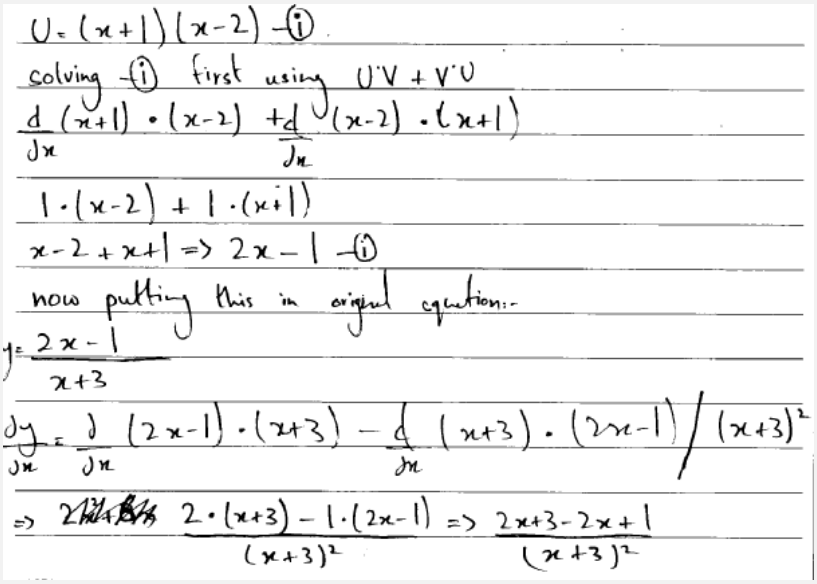
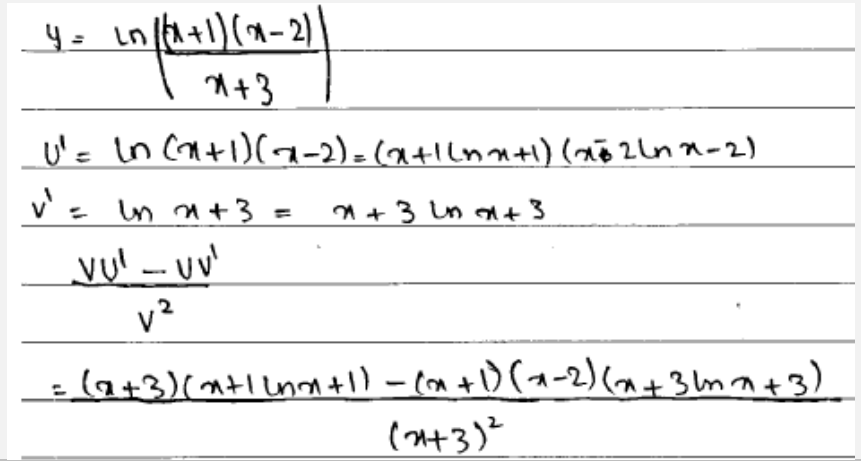
Question No. 2a **Candidates were given the choice to attempt any TWO out of the three questions: 2a, 2b and 2c.**	
Question Text	Using logarithms, differentiate the given function with respect to *x*. $y = \dfrac{(x+1)(x-2)}{x+3}$
SLO No.	13.5.4
SLO Text	Find the derivative of algebraic expressions involving product, quotient and power by using laws of logarithm.
Max Marks	5
Cognitive Level	A
Checking Hints	1 mark for applying logarithms on both sides 1 mark for expansion (2 required) 1 mark for differentiating each function involved 1 mark for getting $\dfrac{dy}{dx} = \dfrac{(x+1)(x-2)}{x+3}\left(\dfrac{1}{x+1}+\dfrac{1}{x+2}-\dfrac{1}{x+3}\right)$
Overall Performance	Relatively lesser number of candidates attempted 2a. The overall performance on this question show that many of the candidates were able to demonstrate a strong grasp of logarithmic differentiation and correctly apply the relevant rules and differentiation techniques. These candidates effectively used the properties of logarithms and differentiation to find the derivative. Conversely, weaker responses show difficulties with recalling and applying the product and quotient formulas for logarithms and differentiating logarithmic functions, leading to incorrect results.
Description of Better Responses	Candidates who performed well on this question accurately applied the properties of logarithms and differentiation. They correctly used logarithmic differentiation to simplify the function before differentiating. By following the proper steps, including the application of logarithmic identities and differentiation rules, these candidates arrived at the correct derivative, demonstrating clear and error-free calculations.
Images of Better Responses	**Image (i)** 

	Image (ii) 
Description of Weaker Responses	In weaker responses, candidates often struggled with recalling the product and quotient rules for logarithms and did not apply these rules correctly during differentiation. Some of the candidates incorrectly used the quotient rule for differentiation instead of the appropriate logarithmic differentiation techniques, resulting in errors in their final answers.
Images of Weaker Response	**Image (i)**  **Image (ii)** 

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:

To enhance candidates' performance in this concept, teachers may consider the following strategies;

Practice Using Interactive Digital Tools:

- **Desmos:** Use Desmos to graph the function and its derivative to help students visualise the relationship between the function and its rate of change. This tool can also help in understanding the effect of logarithmic transformations on the function.
- **Wolfram Alpha:** Utilise Wolfram Alpha for symbolic computation and to verify derivatives and provide step-by-step solutions. This can help students see correct applications of differentiation rules and logarithmic properties.
- **Symbolab:** An online calculator that offers detailed steps for differentiation using logarithms, which can assist students in understanding the process and checking their work.

Real-Life Practical Examples:

- **Finance:** Logarithmic differentiation is used in finance to model compound interest and exponential growth. For example, analysing the growth of investments over time involves differentiating exponential functions.
- **Engineering:** Logarithmic differentiation is employed in engineering fields to solve problems involving rate of change in systems with exponential growth or decay, such as in signal processing or control systems.

Question No. 2b **Candidates were given the choice to attempt any TWO out of the three questions: 2a, 2b and 2c.**	
Question Text	Find $\frac{dy}{dx}$ for the given function. $x = \sin t$, $y = \cos \pi t$
SLO No.	13.3.2
SLO Text	Solve problems related to chain rule.
Max Marks	5
Cognitive Level	A
Checking Hints	1 mark for finding $\frac{dx}{dt} = \cos t$ 2 marks for finding $\frac{dy}{dt} = -\pi \sin \pi t$ (1 mark for function, 1 mark for angle)

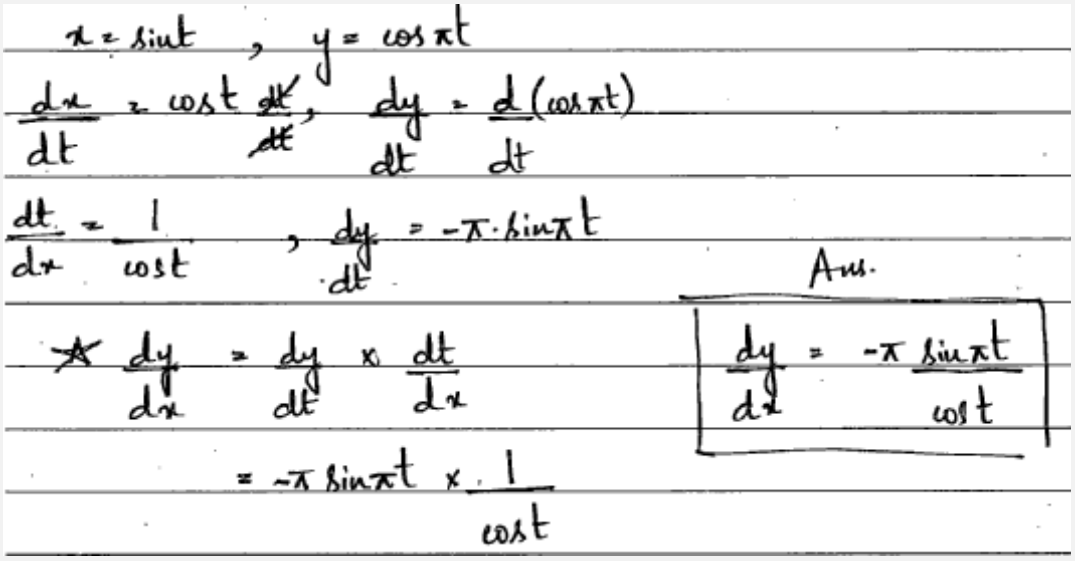
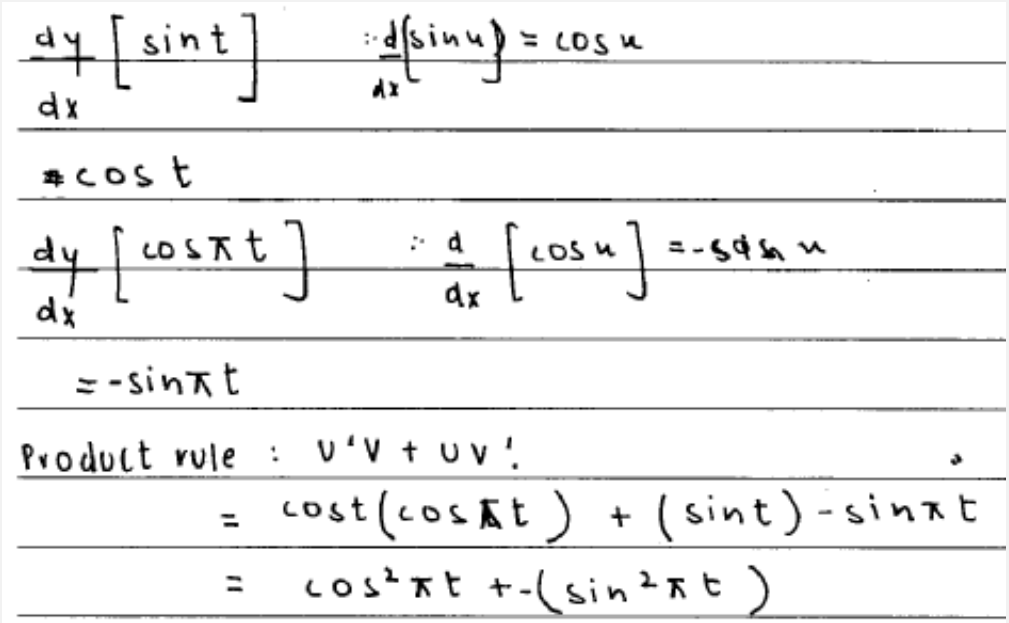
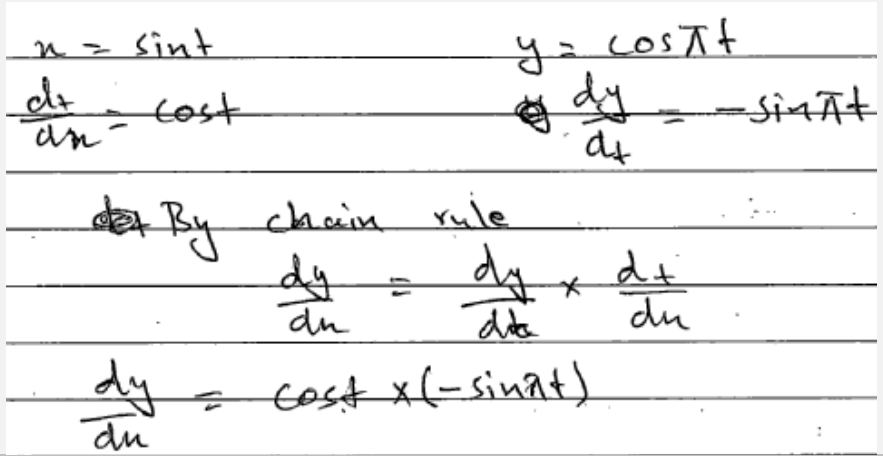
	1 mark for applying the chain rule 1 mark for substitution
Overall Performance	Majority of candidates attempted question 2b. Candidates who understood the differentiation of trigonometric and parametric functions were able to correctly apply the chain rule to find dx/dy. They successfully navigated the complexities of parametric differentiation. However, candidates who struggled often failed to recall or apply the concept of parametric functions and the chain rule, leading to incorrect or incomplete solutions.
Description of Better Responses	Candidates who performed well demonstrated a strong grasp of both parametric differentiation and the chain rule. They correctly differentiated the parametric equations $x=\sin(t)$ and $y=\cos(\pi t)$ with respect to t and then used the chain rule to find dy/dx. These candidates showed clear, accurate mathematical operations and understanding of the relationship between x and y through the parameter (t).
Image of Better Response	
Description of Weaker Responses	Weaker responses indicated a lack of understanding of parametric functions and the chain rule. Many of the candidates differentiated trigonometric functions with respect to a variable other than t and did not use the chain rule correctly. Some responses showed that candidates used product or quotient rules instead of applying the chain rule to parametric functions, leading to incorrect solution.
Images of Weaker Responses	**Image (i)** 

Image (ii)



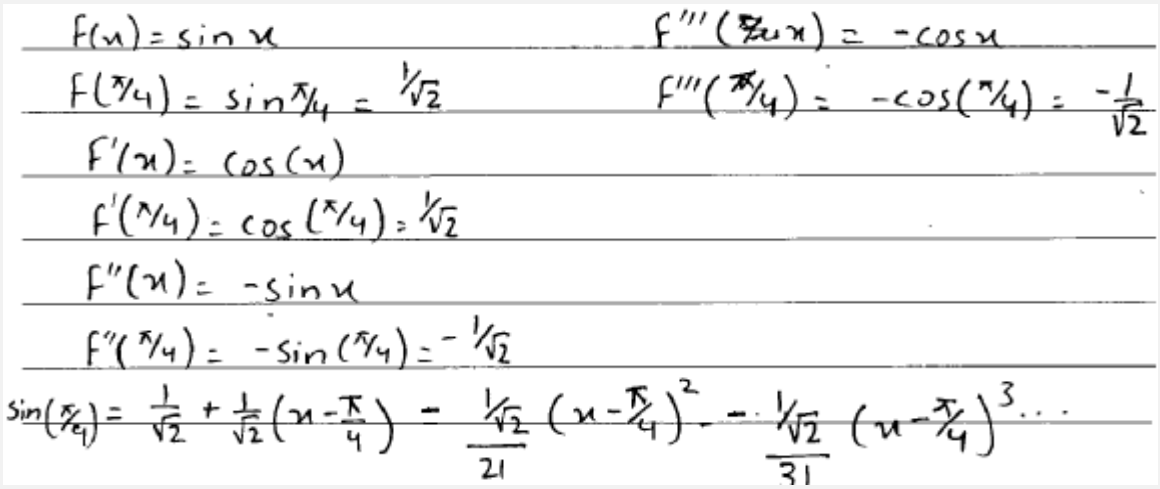
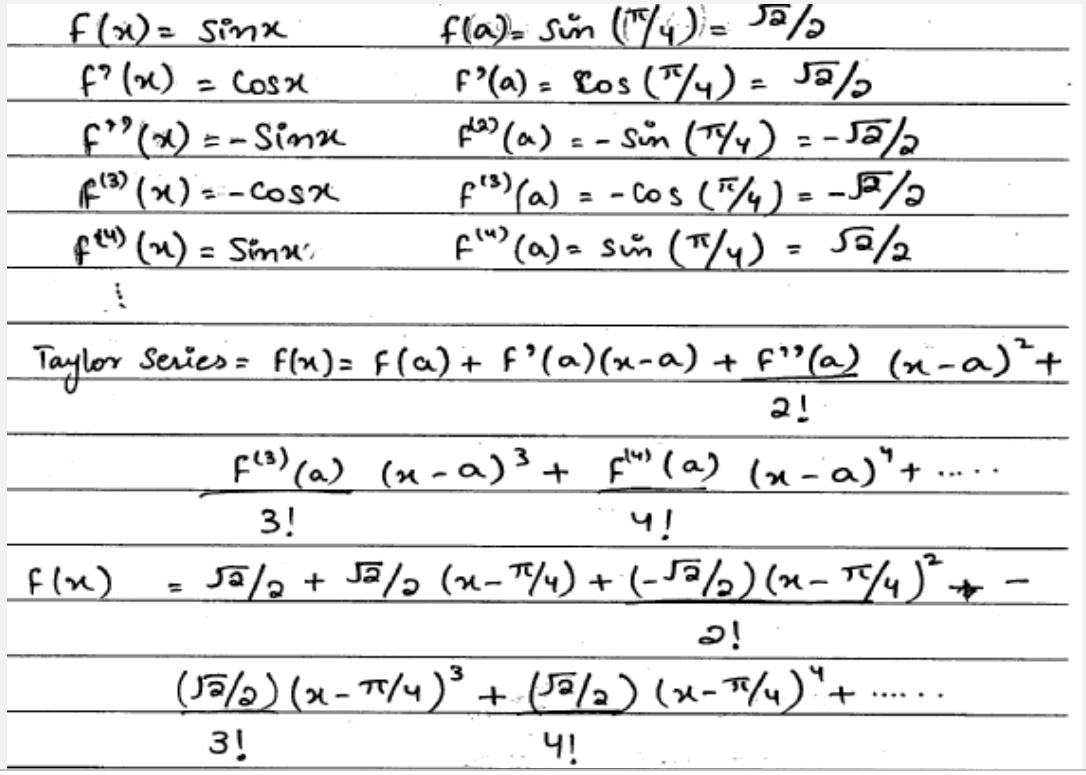
Suggestions for improvement (Highlight all that apply)

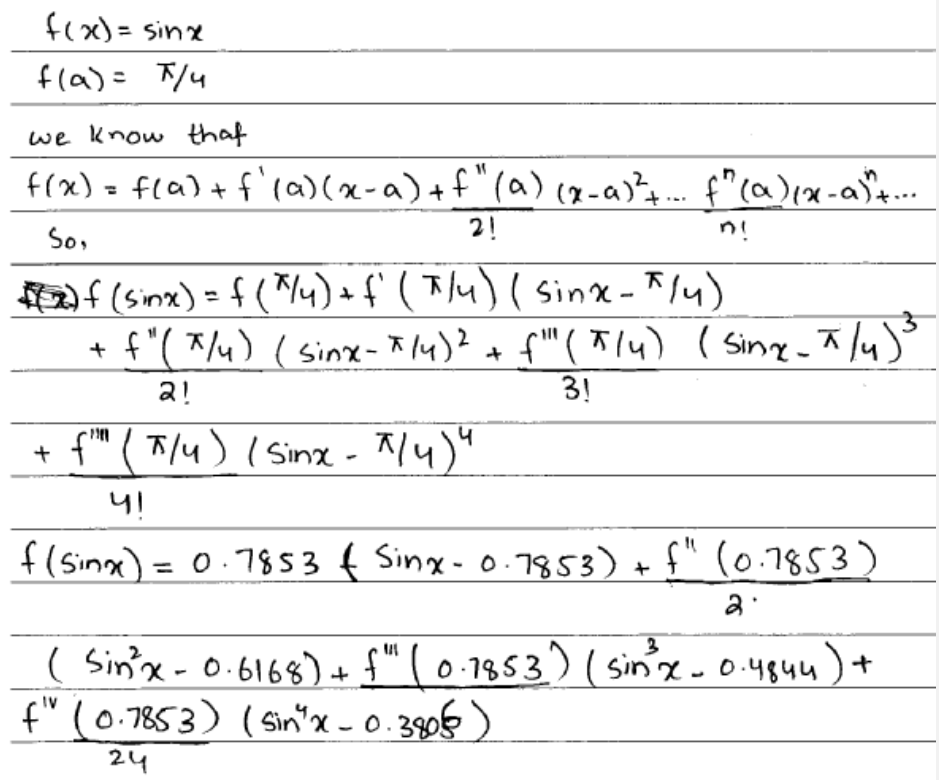
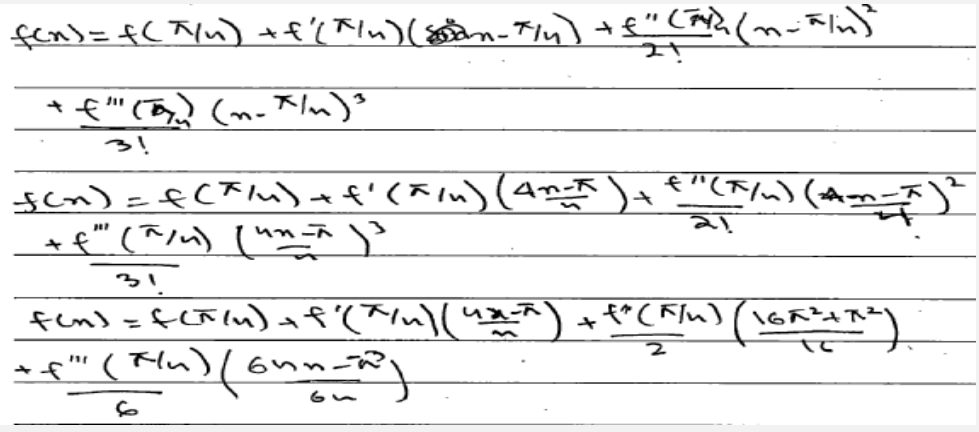
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Any Additional Suggestion:
To enhance overall performance, teachers may consider the following strategies;
Practice Using Interactive Digital Tools:

- **Desmos:** Use Desmos to graph parametric equations and visualise the relationship between *x* and *y*. This can help students understand the effect of changes in the parameter on the function.
- **Wolfram Alpha:** Utilise Wolfram Alpha to compute derivatives for parametric functions, providing step-by-step solutions that illustrate the correct application of the chain rule.
- **Symbolab:** Introduce an online calculator that can handle parametric differentiation, showing detailed steps and explanations for finding *dy*, which can assist students in understanding the process and checking their work.

Question No. 2c **Candidates were given the choice to attempt any TWO out of the three questions: 2a, 2b and 2c.**	
Question Text	Find the Taylor series expansion of the given function about the given value up to four terms. $f(x) = \sin x,\ a = \frac{\pi}{4}$
SLO No.	14.2.2
SLO Text	Apply these theorems to expand sin*x*, cos*x*, tan*x*, a^x, e^x, $\log_a(1+x)$ and $\ln(1+x)$.
Max Marks	5

Cognitive Level	A
Checking Hints	1 mark for finding each derivatives (3 required) 1 mark for substitution in series 1 mark for finding sequence
Overall Performance	The overall performance on this question shows that some of the candidates demonstrated a strong understanding of Taylor series expansion and higher-order differentiation of trigonometric functions, leading to accurate and complete solutions. However, many candidates struggled with recalling the necessary differentiation concepts and correctly applying the Taylor series formula, resulting in incomplete or incorrect expansions.
Description of Better Responses	Candidates who performed well demonstrated a clear understanding of higher-order differentiation of sin (x) and the specific values at $x = \pi/4$. They correctly computed the derivatives, evaluated them at the given point and accurately substituted these values into the Taylor series formula. These candidates showed precise and error-free mathematical operations, producing the correct four-term expansion of the function.
Images of Better Responses	**Image (i)**  $f(x) = \sin x$ $\quad f'''(x) = -\cos x$ $f(\pi/4) = \sin \pi/4 = 1/\sqrt{2}$ $\quad f'''(\pi/4) = -\cos(\pi/4) = -\frac{1}{\sqrt{2}}$ $f'(x) = \cos(x)$ $f'(\pi/4) = \cos(\pi/4) = 1/\sqrt{2}$ $f''(x) = -\sin x$ $f''(\pi/4) = -\sin(\pi/4) = -1/\sqrt{2}$ $\sin(\pi/4) = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}(x - \frac{\pi}{4}) - \frac{1/\sqrt{2}}{2!}(x-\pi/4)^2 - \frac{1/\sqrt{2}}{3!}(x-\pi/4)^3 \ldots$ **Image (ii)**  $f(x) = \sin x$ $\quad f(a) = \sin(\pi/4) = \sqrt{2}/2$ $f'(x) = \cos x$ $\quad f'(a) = \cos(\pi/4) = \sqrt{2}/2$ $f''(x) = -\sin x$ $\quad f^{(2)}(a) = -\sin(\pi/4) = -\sqrt{2}/2$ $f^{(3)}(x) = -\cos x$ $\quad f^{(3)}(a) = -\cos(\pi/4) = -\sqrt{2}/2$ $f^{(4)}(x) = \sin x$ $\quad f^{(4)}(a) = \sin(\pi/4) = \sqrt{2}/2$ $\vdots$ Taylor Series $= f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f^{(3)}(a)}{3!}(x-a)^3 + \frac{f^{(4)}(a)}{4!}(x-a)^4 + \ldots$ $f(x) = \sqrt{2}/2 + \sqrt{2}/2\,(x-\pi/4) + \frac{(-\sqrt{2}/2)}{2!}(x-\pi/4)^2 - \frac{(\sqrt{2}/2)}{3!}(x-\pi/4)^3 + \frac{(\sqrt{2}/2)}{4!}(x-\pi/4)^4 + \ldots$

Description of Weaker Responses	In weaker responses, some of the candidates incorrectly substituted angles instead of evaluating the derivatives at the specified value. Others struggled with placing the values correctly in the Taylor series formula, leading to incorrect or incomplete series expansions. These responses indicate a need for stronger foundational knowledge in both differentiation and series expansion.
Images of Weaker Responses	**Image (i)**  $f(x) = \sin x$ $f(a) = \pi/4$ we know that $f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots \frac{f^n(a)}{n!}(x-a)^n + \dots$ So, $f(\sin x) = f(\pi/4) + f'(\pi/4)(\sin x - \pi/4) + \frac{f''(\pi/4)}{2!}(\sin x - \pi/4)^2 + \frac{f'''(\pi/4)}{3!}(\sin x - \pi/4)^3$ $+ \frac{f''''(\pi/4)}{4!}(\sin x - \pi/4)^4$ $f(\sin x) = 0.7853 + (\sin x - 0.7853) + \frac{f''(0.7853)}{2}$ $(\sin^2 x - 0.6168) + \frac{f'''(0.7853)}{}(\sin^3 x - 0.4844) +$ $\frac{f''''(0.7853)}{24}(\sin^4 x - 0.3805)$ **Image (ii)**  $f(x) = f(\pi/4) + f'(\pi/4)(x - \pi/4) + \frac{f''(\pi/4)}{2!}(x - \pi/4)^2$ $+ \frac{f'''(\pi/4)}{3!}(x - \pi/4)^3$ $f(x) = f(\pi/4) + f'(\pi/4)\left(4x - \frac{\pi}{4}\right) + \frac{f''(\pi/4)}{2!}\left(4x - \frac{\pi}{4}\right)^2$ $+ \frac{f'''(\pi/4)}{3!}\left(4x - \frac{\pi}{4}\right)^3$ $f(x) = f(\pi/4) + f'(\pi/4)\left(4x - \frac{\pi}{4}\right) + \frac{f''(\pi/4)}{2}\left(\frac{16x^2 + \pi^2}{16}\right)$ $+ \frac{f'''(\pi/4)}{6}\left(\frac{64x^3 - \pi^3}{64}\right)$

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

• Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	

Any Additional Suggestion:
By focusing on these strategies, educators can help students better understand and apply the Taylor series expansion;

Practice Using Interactive Digital Tools:

- **Desmos:** Use Desmos to visualise the Taylor series expansion alongside the original function. This can help students understand that the series approximates the function near the given point.
- **Wolfram Alpha:** Utilise Wolfram Alpha to compute Taylor series expansions and verify results. This tool provides step-by-step solutions, helping students learn the correct process and identify any mistakes.
- **Symbolab:** An online calculator that offers detailed steps for finding Taylor series expansions, reinforcing the correct application of differentiation and substitution.

Question No. 3a **Candidates were given the choice to attempt any TWO out of the three questions: 3a, 3b and 3c.**	
Question Text	Using integration by substitution, evaluate $\int \frac{dx}{\sqrt{x^2-9}} = \ln\left(\frac{x}{3} + \frac{\sqrt{x^2-9}}{3}\right) + c$.
SLO No.	16.3.2
SLO Text	Evaluate indefinite integrals using appropriate substitutions.
Max Marks	5
Cognitive Level	A
Checking Hints	1 mark for substituting for x as $x = 3\sec t$ 1 mark for substituting for dx as $dx = 3\sec t \times \tan t\, dt$ 1 mark for simplification and getting $\int \sec t\, dt$ 1 mark for integrating $\int \sec t\, dt$ and writing as $\ln(\sec t + \tan t) + c$ 1 mark for re-substituting and writing the integral in the terms of x as $\ln\left(\frac{x}{3} + \frac{\sqrt{x^2-9}}{3}\right) + c$
Overall Performance	Overall, a lesser number of the candidates attempted this question. Candidates who understood trigonometric substitution for integration were able to correctly solve the integral in terms of x. However, many struggled with the correct substitutions, differentiation and re-substitution, resulting in errors.
Description of Better Responses	Candidates who performed well demonstrated a strong understanding of trigonometric substitution. They correctly identified the appropriate trigonometric substitution, such as x = 3sec (θ) and differentiated it accurately. They then applied this substitution to the integral and simplified it correctly. These candidates accurately integrated the resulting function and performed the re-substitution without any errors, leading to the correct final result.

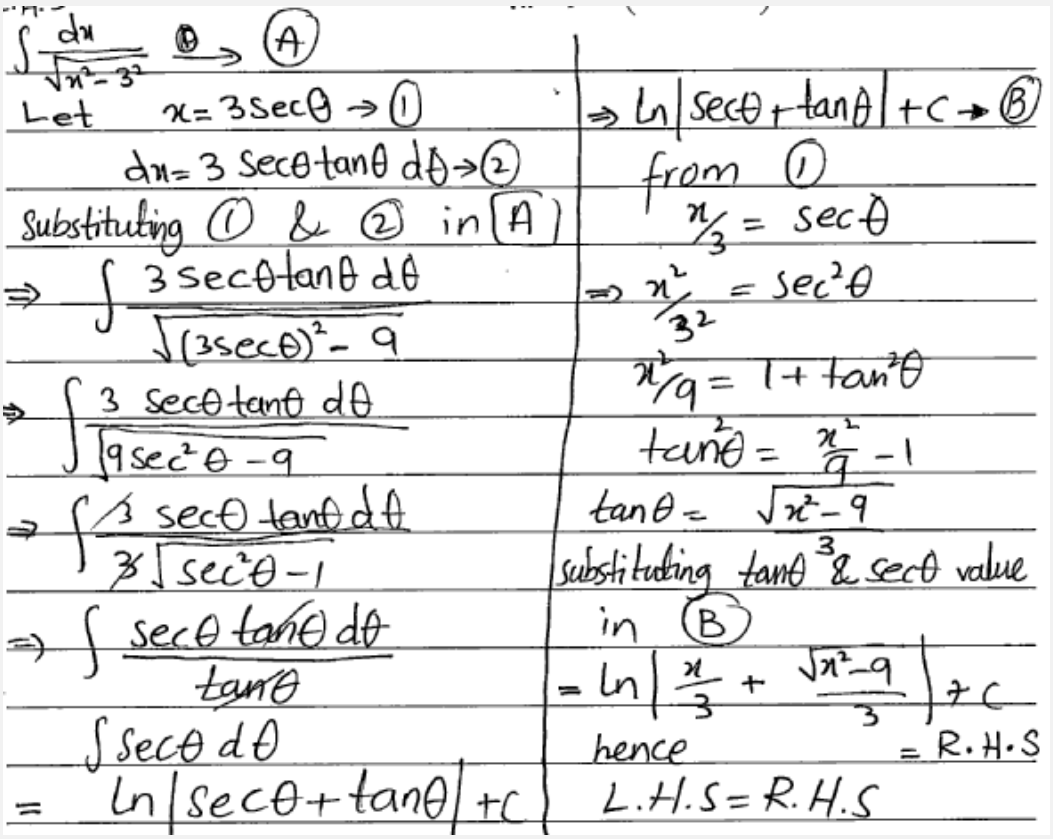
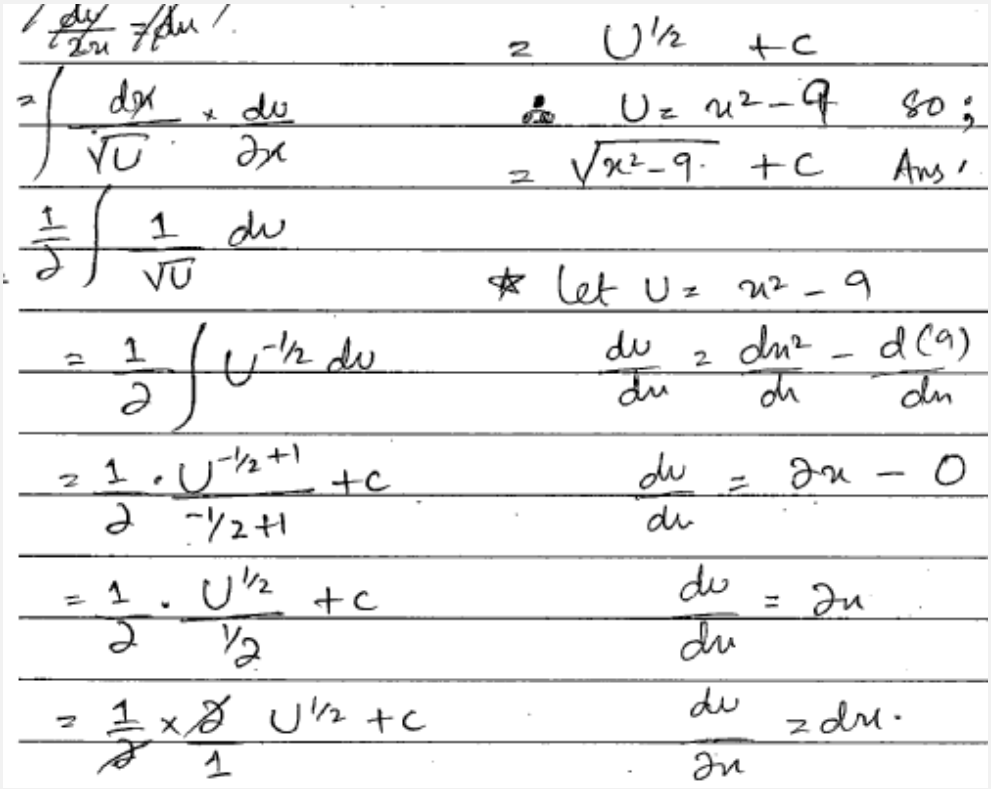
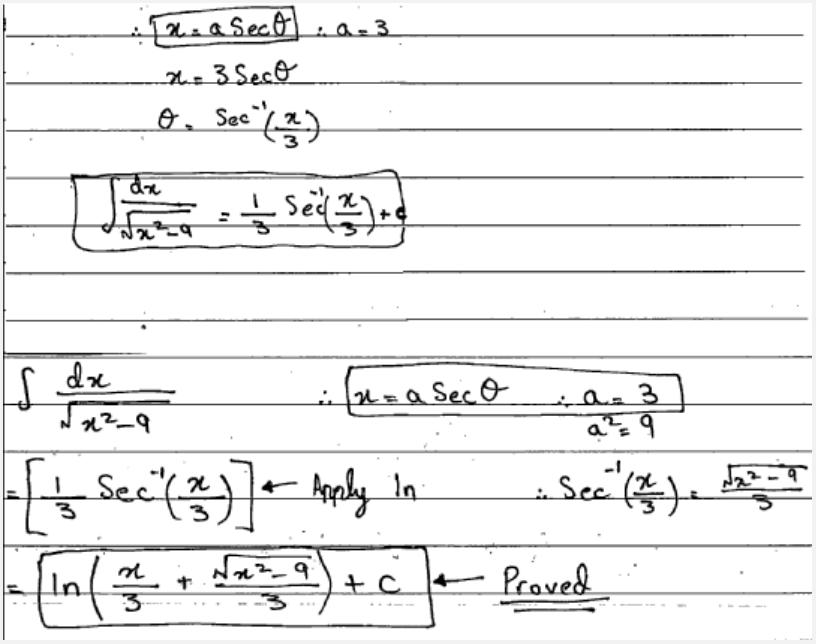
Image of Better Response	
Description of Weaker Responses	Majority of the weaker responses indicated confusion about the correct trigonometric substitution to use. Some of the candidates incorrectly chose algebraic substitutions instead of the necessary trigonometric ones. Even when the correct substitution was chosen, many candidates struggled with differentiating and substituting back into the original integral. Additionally, some of the candidates could not correctly re-substitute to obtain the final result, showing a lack of understanding of the overall process.
Images of Weaker Responses	**Image (i)** 

Image (ii)



Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

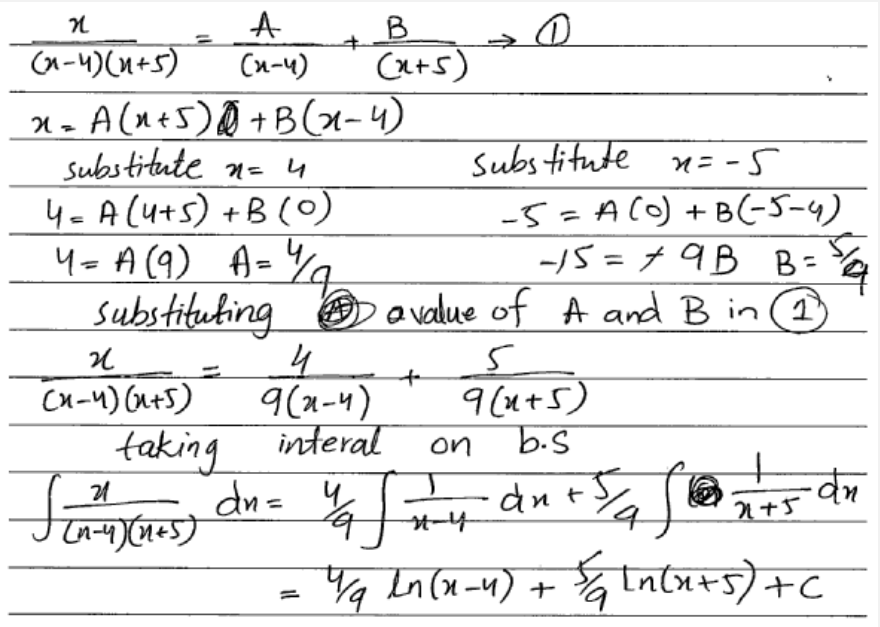
To enhance overall performance, consider the following strategies;

Practice Using Interactive Digital Tools:

- **Desmos:** Use Desmos to visualise the function and its integral. This can help students to understand the geometric interpretation of the integral and the effect of substitutions.
- **Wolfram Alpha:** Utilise Wolfram Alpha to solve integrals step-by-step. This tool provides detailed solutions that help students follow the correct process for trigonometric substitutions and integrations.
- **Symbolab:** An online calculator that offers step-by-step integration, including trigonometric substitutions, helping students see each step and understand the logic behind the substitutions and simplifications.

Targeted Instruction:

- **Focused Remediation:** Provide additional practice problems specifically targeting trigonometric substitutions. Discuss detailed feedback and step-by-step solutions to help students rectify their mistakes.
- **Concept Mapping:** Use concept maps to help students visualise the relationships between different trigonometric identities, substitutions and integration techniques. This can aid in better understanding and application of each method.

Question No. 3b Candidates were given the choice to attempt any TWO out of the three questions: 3a, 3b and 3c.	
Question Text	Apply the method of partial fraction to evaluate the given integral. $\int \frac{x}{(x-4)(x+5)}\,dx$
SLO No.	16.5.1
SLO Text	Find $\int \frac{f(x)}{g(x)}\,dx$ using partial fractions where $f(x)$ and $g(x)$ are algebraic functions such that $g(x) \neq 0$.
Max Marks	5
Cognitive Level	A
Checking Hints	1 mark for writing the fraction as $\frac{x}{(x-4)(x+5)} = \frac{A}{(x-4)} + \frac{B}{(x+5)}$ 1 mark for finding the value of A 1 mark for finding the value of B 1 mark for obtaining correct integral $\frac{4}{9}\ln(x-4)$ 1 mark for obtaining correct integral $\frac{5}{9}\ln(x+5)$
Overall Performance	Majority of the candidates attempted this question. The overall performance on this question shows that while many candidates demonstrated a clear understanding of partial fractions and their integration, some of the candidates struggled with the concept and application of this method, leading to errors in finding and integrating the partial fractions.
Description of Better Responses	Candidates who performed well correctly identified the form of the partial fraction decomposition for the given integrand. They accurately set up the equation $\frac{x}{(x-4)(x+5)} = \frac{A}{(x-4)} + \frac{B}{(x+5)}$ and solved for A and B without any error. These candidates then integrated the resulting partial fractions correctly, demonstrating a strong grasp of the required mathematical operations .
Image of Better Response	

$$\frac{x}{(x-4)(x+5)} = \frac{A}{(x-4)} + \frac{B}{(x+5)} \rightarrow (1)$$

$$x = A(x+5) + B(x-4)$$

Substitute $x = 4$: $4 = A(4+5) + B(0)$, $4 = A(9)$, $A = \frac{4}{9}$

Substitute $x = -5$: $-5 = A(0) + B(-5-4)$, $-15 = -9B$, $B = \frac{5}{9}$

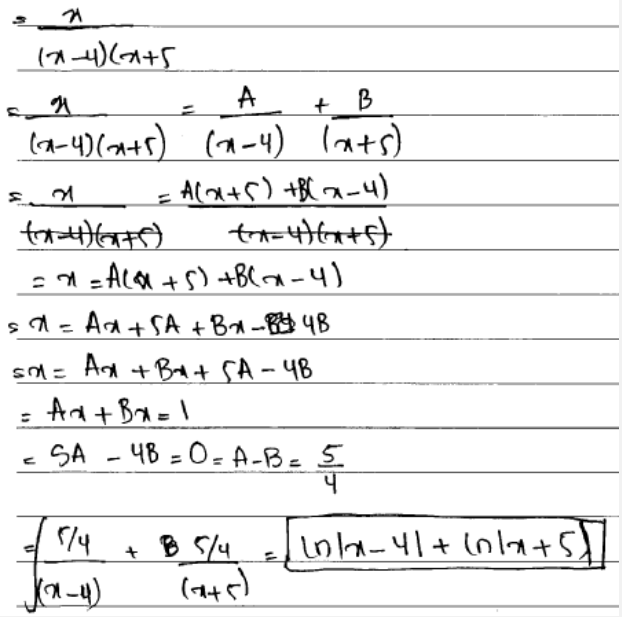
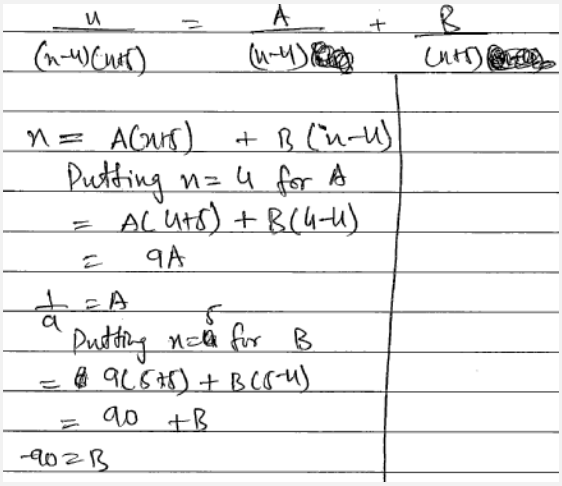
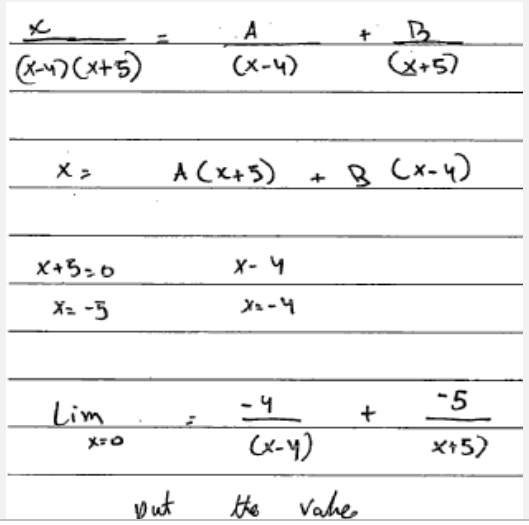
Substituting A value of A and B in (1)

$$\frac{x}{(x-4)(x+5)} = \frac{4}{9(x-4)} + \frac{5}{9(x+5)}$$

taking interal on b.s

$$\int \frac{x}{(x-4)(x+5)}\,dx = \frac{4}{9}\int \frac{1}{x-4}\,dx + \frac{5}{9}\int \frac{1}{x+5}\,dx$$

$$= \frac{4}{9}\ln(x-4) + \frac{5}{9}\ln(x+5) + C$$

Description of Weaker Responses	Weaker responses showed that while some of the candidates could identify the correct form for the partial fractions, they often struggled to find the correct values for the unknowns *A* and *B*. Errors in algebraic manipulation were common, leading to incorrect partial fractions. Additionally, some of the candidates failed to integrate the resulting fractions correctly, indicating a lack of understanding of the integration process for partial fractions.
Images of Weaker Responses	**Image (i)**  **Image (ii)**  **Image (iii)** 

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

To enhance overall performance, teachers may consider the following strategies;

Practice Using Interactive Digital Tools:

- **Wolfram Alpha:** Use Wolfram Alpha to perform partial fraction decompositions and integrations step-by-step. This can help students see the correct values for the unknowns and understand the integration process.
- **Symbolab:** An online calculator that offers detailed steps for partial fraction decomposition and integration, helping students follow the correct method and check their work.
- **Geogebra:** Utilise GeoGebra to visualise the integrand and its decomposition. This tool can help students understand the relationship between the original function and its partial fractions.

Targeted Instruction:

- **Focused Practice Problems:** Provide additional practice problems specifically on partial fractions and their integration. Discuss detailed solutions and step-by-step explanations to help students understand the correct process.

Question No. 3c **Candidates were given the choice to attempt any TWO out of the three questions: 3a, 3b and 3c.**	
Question Text	An object is dropped from the top of a tower, vertically downwards, under the influence of gravitational force. Taking gravitational acceleration as 10 m/s^2, find the i. expression for velocity v after t seconds of falling. ii. velocity after 2 seconds of the fall. iii. hence, find the expression for displacement.
SLO No.	16.7.3
SLO Text	Solve word problems related to differentiation and integration, e.g. finding displacement from velocity, etc.
Max Marks	5
Cognitive Level	A

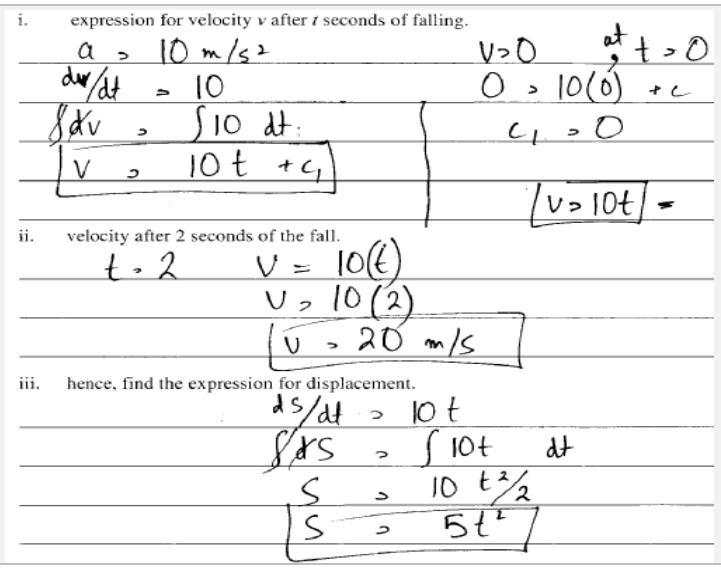
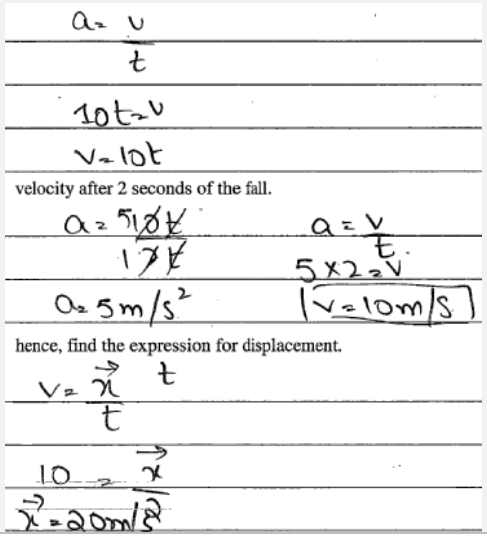
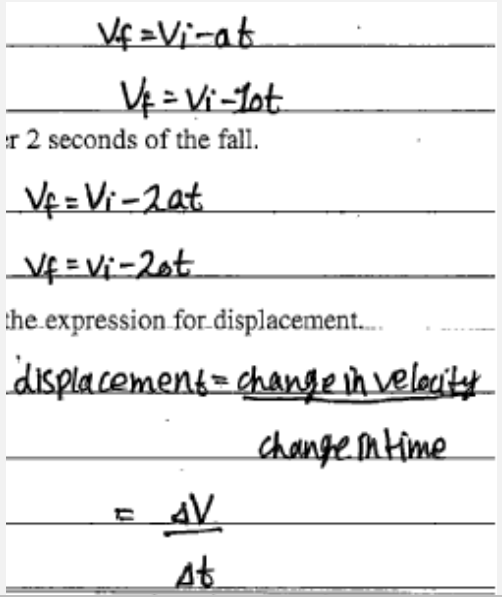
Checking Hints	i. 1 mark for taking $a = \frac{dv}{dt} = 10$ 1 mark for integrating both sides and writing $v = 10t + C$ 1 mark for finding the value of c and writing $v = 10t$ 1 mark for finding the value of v after 2 seconds of falling ii. 1 mark for obtaining the expression for displacement as $S = 5t^2$
Overall Performance	A considerable number of candidates attempted question 3c. The performance on this question shows that some of the candidates demonstrated a strong understanding of the application of calculus in Physics, particularly in relating acceleration, velocity and displacement. However, many struggled with recalling these concepts and applying them correctly to solve the given problem.
Description of Better Responses	Candidates who performed well were familiar with the application of calculus in daily life, particularly in Physics and Mechanics. They correctly used the relationship between acceleration, velocity and displacement to find the required results. These candidates accurately integrated the given acceleration to find the velocity expression, evaluated the velocity at 2 seconds and then integrated the velocity to find the displacement expression. Their solutions were free from mathematical errors.
Image of Better Response	 i. expression for velocity v after t seconds of falling. $a = 10\ m/s^2$ $dv/dt = 10$ $\int dv = \int 10\ dt$ $v = 10t + c_1$ $v = 0$ at $t = 0$ $0 = 10(0) + c$ $c_1 = 0$ $v = 10t$ ii. velocity after 2 seconds of the fall. $t = 2$ $v = 10(t)$ $v = 10(2)$ $v = 20\ m/s$ iii. hence, find the expression for displacement. $ds/dt = 10t$ $\int ds = \int 10t\ dt$ $s = 10t^2/2$ $s = 5t^2$
Description of Weaker Responses	Weaker responses indicated that candidates have struggled with recalling the application of calculus in Physics and Mechanics. Many failed to track the relationship between acceleration, velocity and displacement, made errors in the integration process or did not apply the correct formulas. In addition, few candidates also could not correctly evaluate the velocity after 2 seconds or derive the displacement expression from the velocity.
Images of Weaker Responses	**Image (i)**  $a = v/t$ $10t = v$ $v = 10t$ velocity after 2 seconds of the fall. $a = 5t/t$ $a = 5\ m/s^2$ $a = v/t$ $5 \times 2 = v$ $v = 10\ m/s$ hence, find the expression for displacement. $v = \vec{x}/t$ $10 = \vec{x}/t$ $\vec{x} = 20\ m/s$

Image (ii)



Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

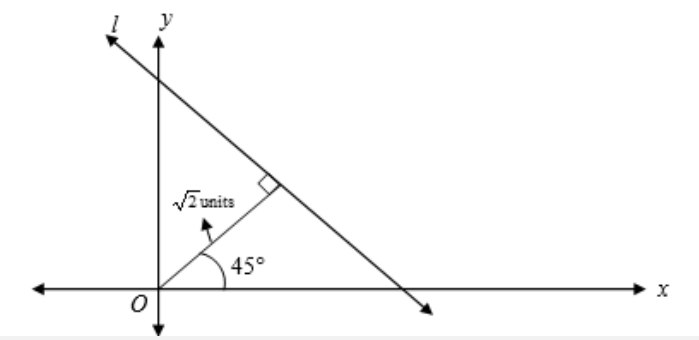
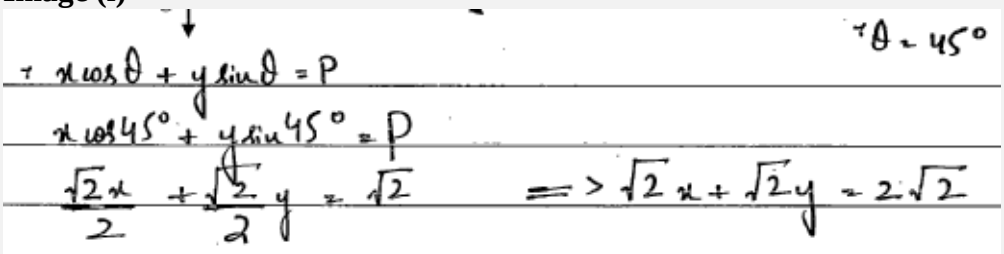
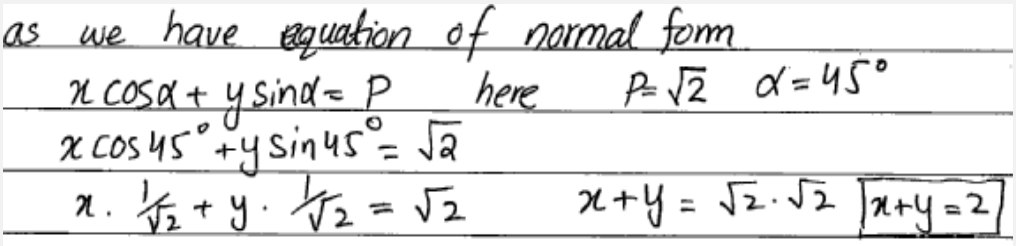
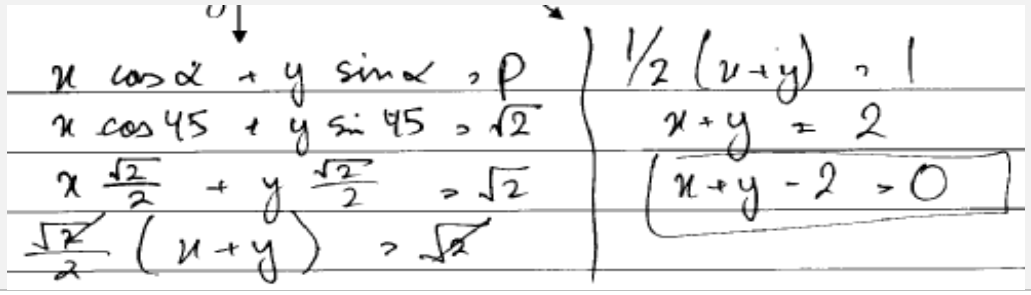
To enhance overall performance, teacher may consider the following strategies;

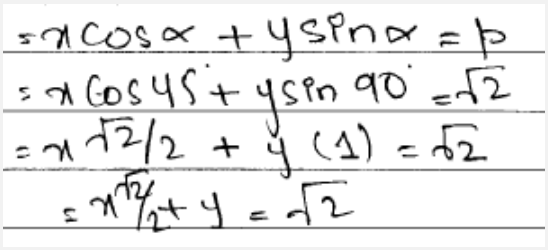
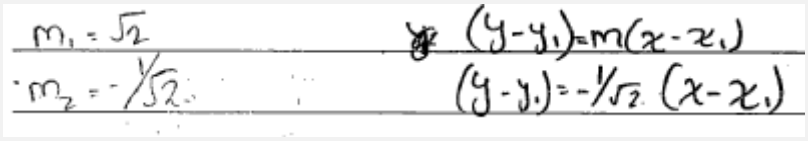
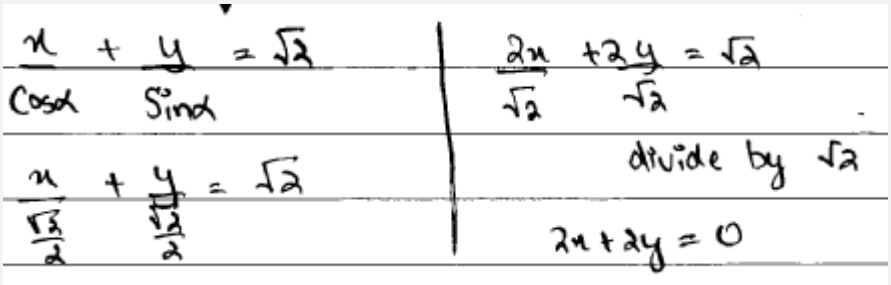
Practice Using Interactive Digital Tools:

- **Desmos:** Use Desmos to graph velocity and displacement functions. This visualisation can help students understand the relationship between acceleration, velocity and displacement.
- **Wolfram Alpha:** Utilise Wolfram Alpha to perform integrations and check the correctness of velocity and displacement expressions. This tool provides step-by-step solutions and immediate feedback.

Targeted Instruction:

- **Problem-Solving Sessions:** Conduct focused sessions where students work on physics-related calculus problems. Provide detailed solutions and explanations to reinforce the concepts.
- **Real-Life Examples:** Use real-life examples and applications of calculus in Physics to help students relate to the concepts. Discuss scenarios like free-falling objects, projectile motion and other problems related to mechanics to demonstrate the practical use of calculus.

Question No. 4i	
Question Text	Find the equation of the line l in the given situation by using the normal form. 
SLO No.	17.4.4
SLO Text	Find the equation of straight line when different situations are given.
Max Marks	2
Cognitive Level	A
Checking Hints	1 mark for employing the normal form 1 mark for substituting the values
Overall Performance	The performance on this question depicts that majority of the candidates have demonstrated a strong understanding of the normal form of the equation of a straight line and applied it correctly. However, some struggled with recognising and correctly using the normal form, leading to incorrect equations.
Description of Better Responses	Candidates, who performed well have recognised the normal form of the equation of a straight line, which is $x\cos\theta + y\sin\theta = p$. They correctly substituted the given angle $\theta = 45°$ and the perpendicular distance $p = 2$ into the equation. These candidates accurately performed the mathematical operations, resulting in the correct equation of the line.
Images of Better Responses	**Image (i)**  **Image (ii)**  **Image (iii)** 

Description of Weaker Responses	Weaker responses struggled with recognising the normal form of the equation of a straight line. Out of these, many could not correctly substitute the given angle and perpendicular distance into the equation. Common errors include incorrect placements of the angle and distance, leading to incorrect or incomplete equations. These responses show a lack of understanding of the normal form and its application.
Images of Weaker Responses	**Image (i)**  **Image (ii)**  **Image (iii)** 

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

To enhance overall performance, consider the following strategies;

Practice Using Interactive Digital Tools:

- **GeoGebra:** Use GeoGebra to visualise the equation of a line given in different forms, including the normal form. This helps students see the relationship between the angle, distance and the resulting line.

- **Desmos:** Utilise Desmos to graph lines using different forms of equations. Students can input the normal form and see the effect of changing the angle and distance on the line's position.

Targeted Instruction:

- **Worked Examples:** Provide step-by-step worked examples of finding the equation of a line using the normal form. Highlight the correct placement of angle and distance and the resulting mathematical operations.
- **Practice Problems:** Offer practice problems specifically focused on the normal form of the equation of a line. Include a variety of angles and distances to reinforce the concept and its practical application.
- **Conceptual Understanding:** Emphasise the geometric interpretation of the normal form, showing that the angle and perpendicular distance define the line's orientation and position relative to the origin.

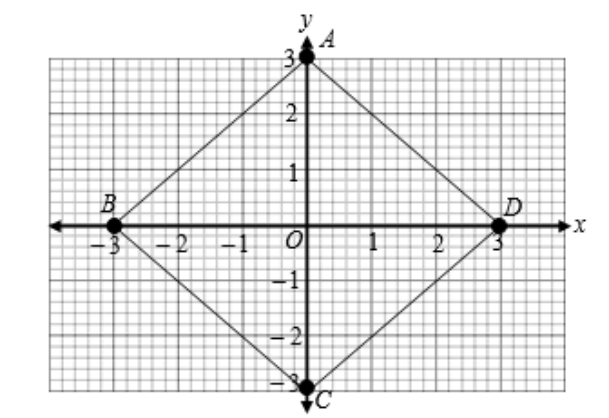
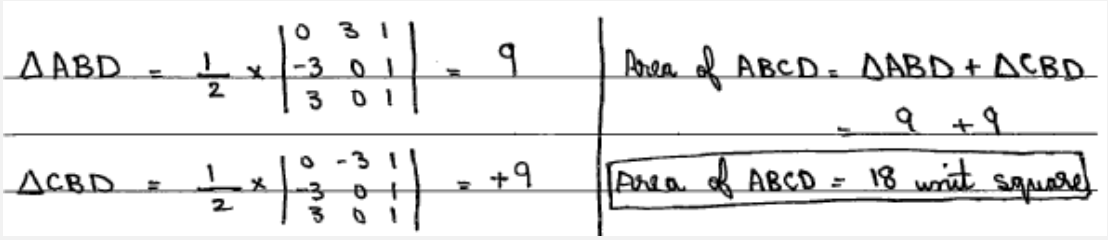
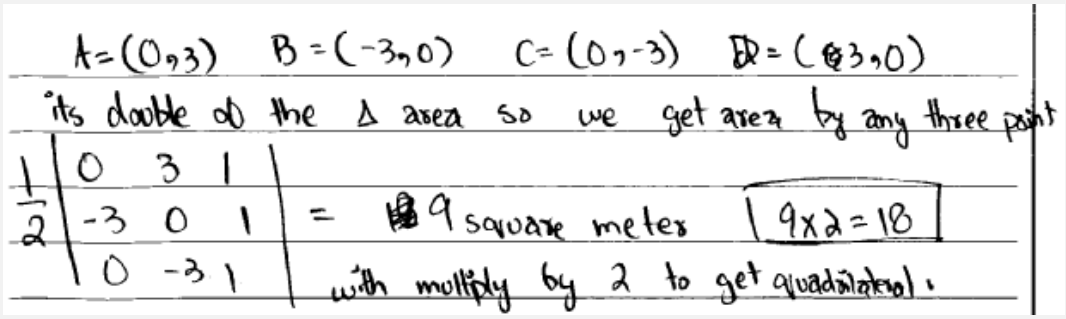
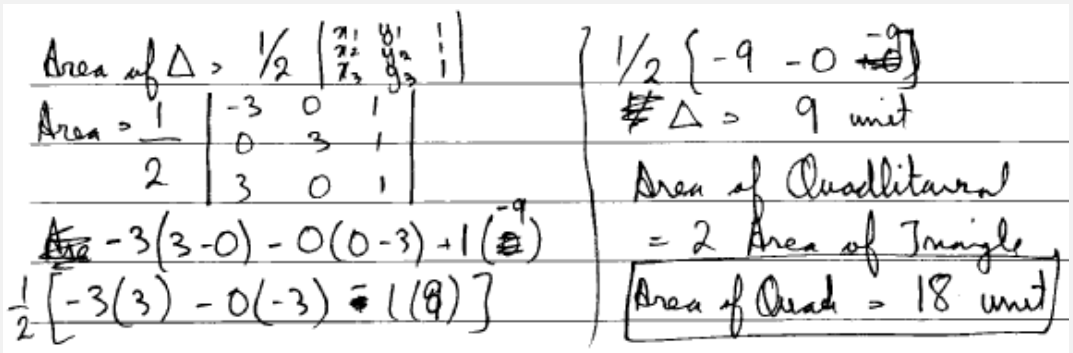
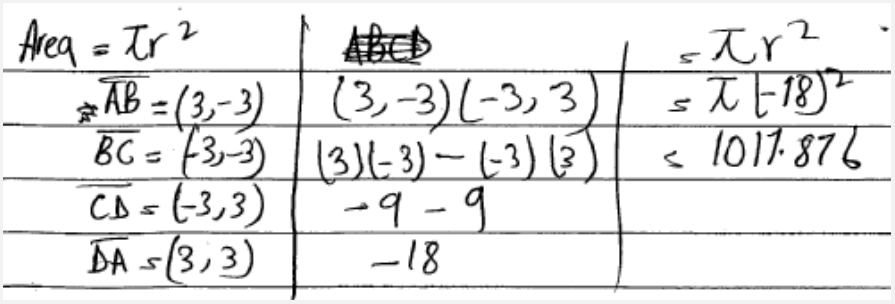
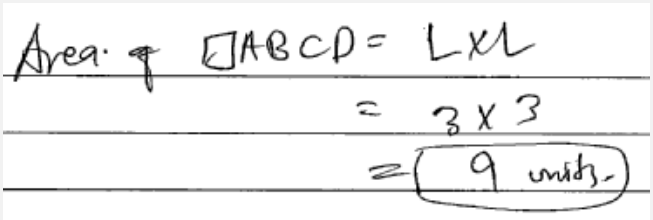
Question No. 4ii	
Question Text	Find the area of the quadrilateral *ABCD* with the given vertices. 
SLO No.	17.8.1
SLO Text	Find the area of a triangular region and hence quadrilateral region whose vertices are given.
Max Marks	2
Cognitive Level	A
Checking Hints	1 mark for area of a triangle in determinant form 1 mark for adding the two area to get the area of the quadrilateral
Overall Performance	The overall performance on this question depicts that the candidates who understood the techniques for finding the area of a quadrilateral by dividing it into triangles performed well, while few of the candidates struggled with correctly identifying and applying the appropriate methods.
Description of Better Responses	Candidates who performed well demonstrated the use of a technique for finding the area of a quadrilateral, i.e., by dividing it into two triangles. They correctly identified that the area of the quadrilateral can be found by summing the areas of these two triangles. Some of the candidates also recognised the specific type of quadrilateral, such as a rectangle, and used this understanding to simplify the calculation, doubling the area of one triangle to find the total area of rectangle. These candidates used clear logical steps in their solutions, showing the accuracy in mathematical operations and strong understanding of the concepts.
Images of Better Responses	**Image (i)** 

	Image (ii)  **Image (iii)** 
Description of Weaker Responses	Weaker responses struggled with recognising the technique to find the area of a quadrilateral. Many incorrectly identified the quadrilateral as a square or a circle and applied inappropriate formulas. These candidates showed confusion in dividing the quadrilateral into triangles or miscalculated the areas of the triangles, leading to incorrect results. Their responses reveal a lack of understanding of the geometric properties and methods required for solving the given problem.
Images of Weaker Responses	**Image (i)**  **Image (ii)** 

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide)	• Story Board • Cause and Effect • Fish and Bone	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform

• Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

To enhance overall performance, consider the following strategies;

Practice Using Interactive Digital Tools:

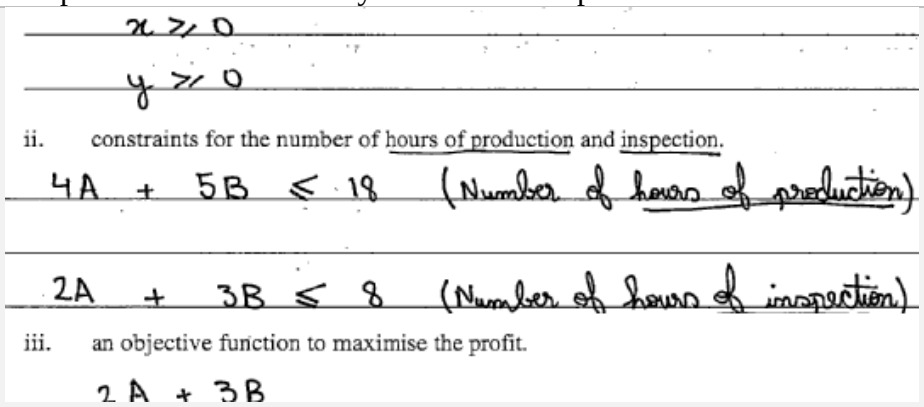
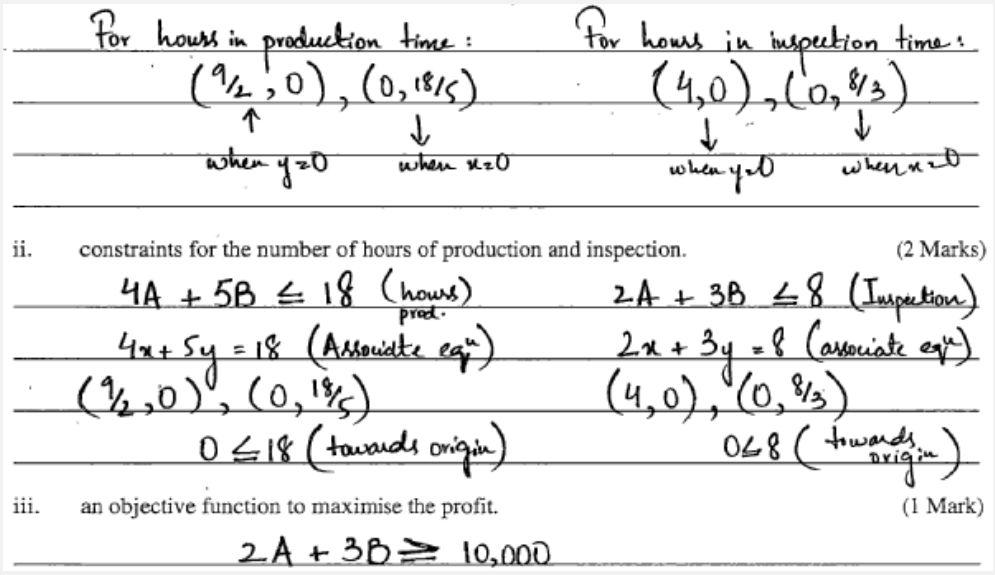
- **GeoGebra:** Use GeoGebra to visualise and manipulate quadrilaterals. Students can see that dividing a quadrilateral into triangles helps in calculating the area. This tool can also dynamically show the area calculations based on vertex positions.
- **Desmos:** Utilise Desmos to plot the vertices of the quadrilateral and compute the areas of the triangles formed. This helps in visualising the problem and understanding the geometric relationships.

Targeted Instruction:

- **Worked Examples:** Provide step-by-step worked examples of finding the area of various quadrilaterals by dividing them into triangles. Highlight the correct identification of triangles and the calculation process.
- **Practice Problems:** Offer practice problems with different types of quadrilaterals and their vertices. Include solutions that demonstrate dividing the quadrilateral into triangles and summing their areas.
- **Geometric Understanding:** Emphasise the importance of understanding geometric properties and their application in problem-solving. Discuss the characteristics of different quadrilaterals and the appropriate methods for calculating their areas.

Question No. 5

Question Text	The given table shows the production time, inspection time and profit per unit of the two manufactured cars *A* and *B*. **Types** | **Production Time in Hours (Per Unit)** | **Inspection Time in Hours (Per Unit)** | **Profit in Rs 10,000 (Per Unit)** ***A*** | 4 | 2 | 2 ***B*** | 5 | 3 | 3 **Maximum** | 18 | 8 | Based on this linear programing problem, construct i. constraints for the values of unknowns (x and y). ii. constraints for the number of hours of production and inspection. iii. an objective function to maximise the profit.
SLO No.	18.4.3
SLO Text	Solve simple linear programing problems.
Max Marks	4
Cognitive Level	A
Checking Hints	i. 1 mark for the general constraint ii. 1 mark for each constraint (2 required) iii. 1 mark for the correct objective function

Overall Performance	Overall, majority of the candidates solved this question partially correct. However, some of the candidates attempted this question correctly and completely scoring full marks. Candidates who understood the terms and methods of linear programming could successfully translate the real-life problem into constraints and an objective function. However, candidates also struggled with the understanding of concepts and the required mathematical formulation.
Description of Better Responses	Candidates who performed well demonstrated a strong understanding of the terms of linear programming and their application in daily life. They correctly identified the variables x (units of car A) and y (units of car B) and formulated the constraints for production and inspection times. These candidates accurately wrote the objective function to maximise profit and performed the necessary mathematical operations without error.
Image of Better Response	
Description of Weaker Responses	The majority of candidates struggled to identify the correct profit function some represented it in inequality form, while others did not write it at all. Many candidates also failed to write the general constraints. However, some of the candidates correctly wrote the constraints for the number of hours of production and inspection (part ii of the question) but mistakenly converted these constraints into equations to find corner points, which was not required. These responses indicate a lack of clear understanding of linear programming terms and their application in real-life scenarios
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide)	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping	• Past paper questions • Discussion on E-Marking Notes

• Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:
To enhance overall performance, consider the following strategies;

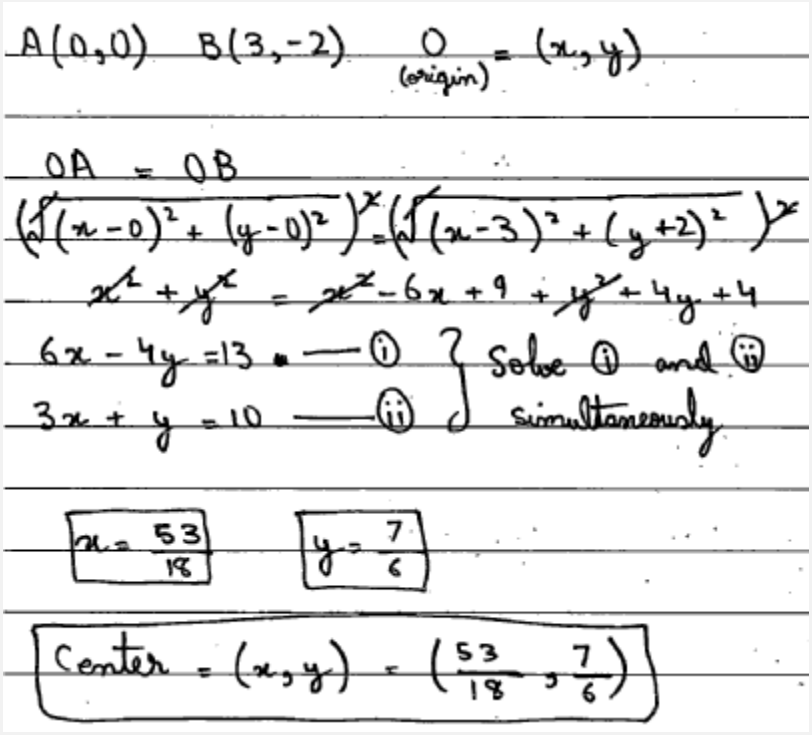
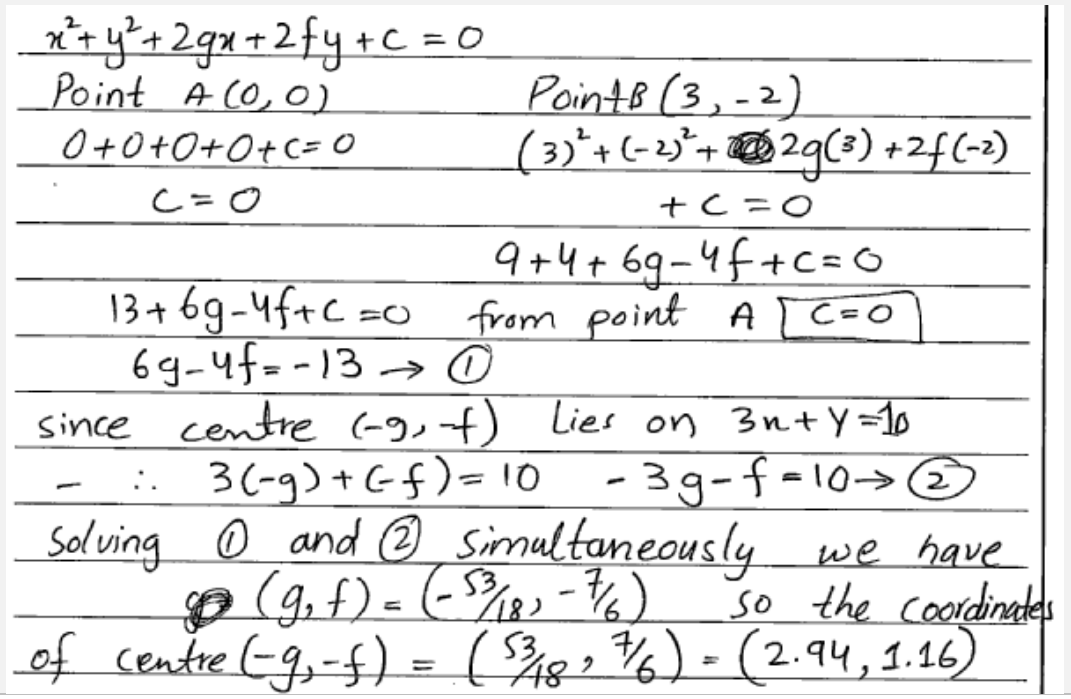
Practice Using Interactive Digital Tools:

- **Desmos:** Use Desmos to visualise the feasible region defined by the constraints and to identify the optimal solution. This can help students understand the graphical interpretation of linear programming problems.
- **GeoGebra:** Utilise GeoGebra to plot the constraints and the objective function. This interactive tool can show the effect of changes in constraints and objective function on the solution.

Targeted Instruction:

- **Worked Examples:** Provide step-by-step worked examples of translating real-life problems into linear programming formulations. Highlight the process of defining variables, writing constraints and forming the objective function.
- **Practice Problems:** Offer a variety of linear programming problems with different contexts. Include detailed solutions to reinforce the concepts and methods.
- **Conceptual Understanding:** Emphasise the importance of understanding the underlying concepts of linear programming. Discuss different ways in which real-life scenarios can be modeled and solved using linear programming techniques.

Question No. 6	
Question Text	Find the centre of the circle passing through the points (0, 0) and (3, –2) and having centre at the line $3x + y = 10$.
SLO No.	19.3.4a
SLO Text	Find the equations of a tangent and a normal to a circle a) at a point.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for making equation 1 in any form (h, k or x, y) 1 mark for comparing equation 2 and 3 1 mark for finding h 1 mark for finding k
Overall Performance	Overall, candidates struggled in solving this question. Most of the candidates could either solve this question partially or could not solve at all. Candidates who understood the concepts of circle equations and the methods to find the centre were able to solve the given problem accurately. However, candidates who struggled with the application of these concepts could not perform well.
Description of Better Responses	Candidates who performed well demonstrated familiarity with the formulae for the radius and the standard and general equations of a circle. They correctly identified the need to use the given points and the condition that the centre lies on the line $3x + y = 10$. These candidates either used the distance formula to find the radius or substituted the points into the standard

	or general circle equation and solved the resulting equations simultaneously with the given line equation. They accurately found the centre without any errors in mathematical operations.
Images of Better Responses	**Image (i)**  A(0,0) B(3,-2) O (origin) = (x, y) OA = OB $(\sqrt{(x-0)^2+(y-0)^2})^2 = (\sqrt{(x-3)^2+(y+2)^2})^2$ $x^2+y^2 = x^2-6x+9+y^2+4y+4$ $6x-4y=13$ ——— (i) } Solve (i) and (ii) $3x+y=10$ ——— (ii) } simultaneously $x=\frac{53}{18}$ $y=\frac{7}{6}$ Center $=(x,y)=(\frac{53}{18},\frac{7}{6})$ **Image (ii)**  $x^2+y^2+2gx+2fy+c=0$ Point A (0, 0) Point B (3, -2) $0+0+0+0+c=0$ $(3)^2+(-2)^2+2g(3)+2f(-2)+c=0$ $c=0$ $9+4+6g-4f+c=0$ $13+6g-4f+c=0$ from point A $c=0$ $6g-4f=-13 \rightarrow (1)$ since centre $(-g,-f)$ lies on $3x+y=10$ $\therefore 3(-g)+(-f)=10$ $-3g-f=10 \rightarrow (2)$ Solving (1) and (2) simultaneously we have $(g,f)=(-\frac{53}{18},-\frac{7}{6})$ so the coordinates of centre $(-g,-f)=(\frac{53}{18},\frac{7}{6})=(2.94,1.16)$
Description of Weaker Responses	In weaker responses, many candidates struggled to correctly use the given points in the standard or general equation of the circle. Some incorrectly substituted the points as (3-*h*, -2-*k*) instead of (3, -2), resulting in incorrect values of *h* and *k* when comparing them with the given equation of the line. Others, despite correctly substituting the points and forming the second equation, did not compare it with the given equation of the line. Additionally, some of the candidates used the given points and the origin to find the slope, created the equation of the line using these points and the slope, and applied midpoints to find the values of *h* and *k*. This led to incorrect or incomplete solutions, indicating a need for reinforcement of the respective concepts and techniques

Images of Weaker Responses

Image (i)

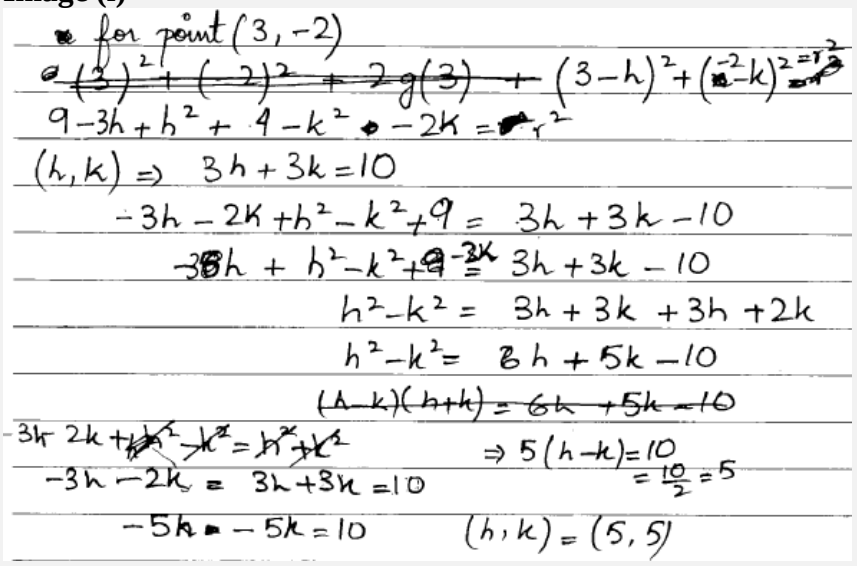


Image (ii)

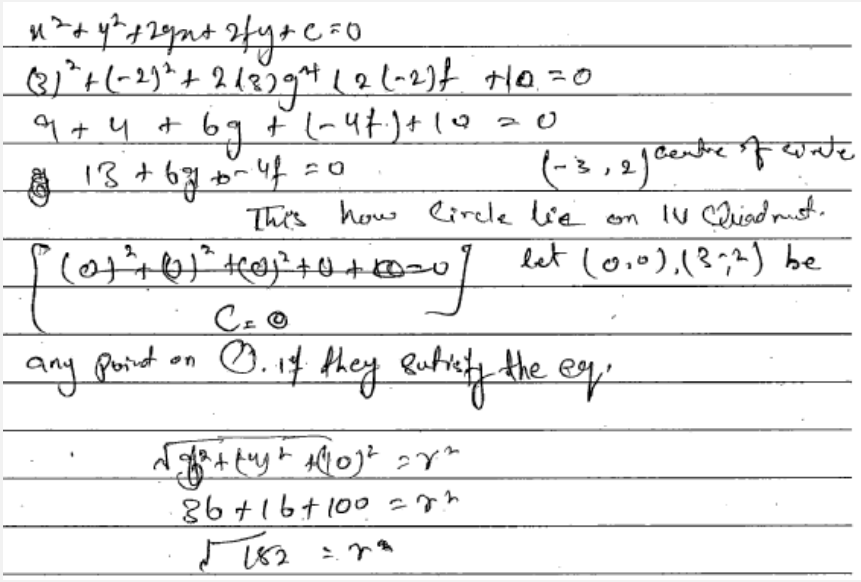
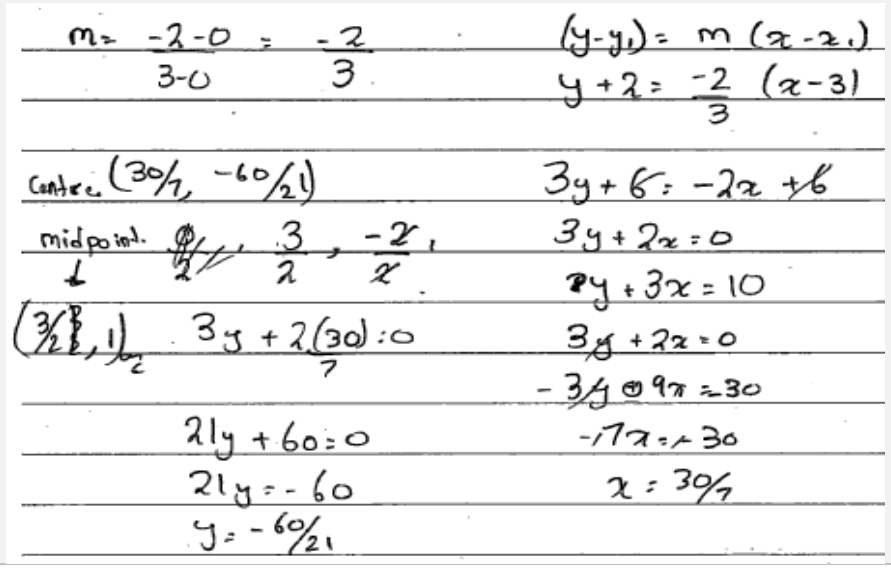


Image (iii)



Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

To enhance overall performance, teachers may consider the following strategies;

Practice Using Interactive Digital Tools:

- **GeoGebra:** Use GeoGebra to visualise the circle passing through given points and the line where the centre lies. This can help students to understand the geometric relationships and the algebraic methods needed to find the centre.
- **Desmos:** Utilise Desmos to plot the points, the circle and the line. This tool allows students to find the centre of the circle by intersecting the line with the locus of possible circle centres.

Targeted Instruction:

- **Worked Examples:** Provide detailed examples that show the step-by-step process of finding the centre of a circle under certain conditions. Highlight the use of the distance formula, the circle equations and solving simultaneous equations.
- **Practice Problems:** Offer a variety of practice problems with different configurations of points and lines. Include detailed solutions to reinforce the concepts and techniques.
- **Conceptual Understanding:** Emphasise the importance of understanding the geometric properties of circles and the algebraic methods to find their centres. Discuss the impact of different given conditions (such as points on the circle and lines where the centre lies) on solving the problem.

Question No. 7a **Candidates were given the choice to attempt any TWO out of the three questions: 7a, 7b, and 7c.**	
Question Text	Find the equation of the normal to the parabola $y^2 = -8x$ at (2, 2) in the standard form.
SLO No.	20.3.2
SLO Text	Find the equation of a tangent and a normal to a parabola (i). at a point.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for taking the derivative OR 1 mark for finding the value of a 1 mark for finding the slope of the tangent OR 1 mark for applying the correct formula 1 mark for finding the slope of the normal OR 1 mark for correct substitution in the formula 1 mark for writing the equation in standard form
Overall Performance	Overall, though lesser number of candidates attempted this question, and majority of the candidates attempting this question were not able to score full marks. Most of the candidates

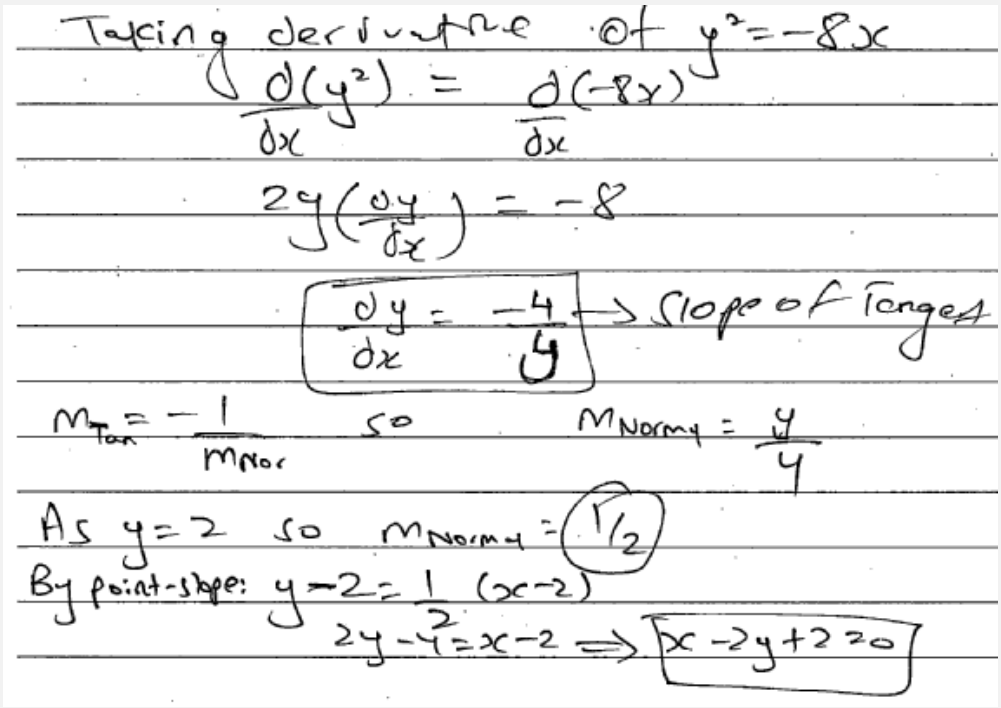
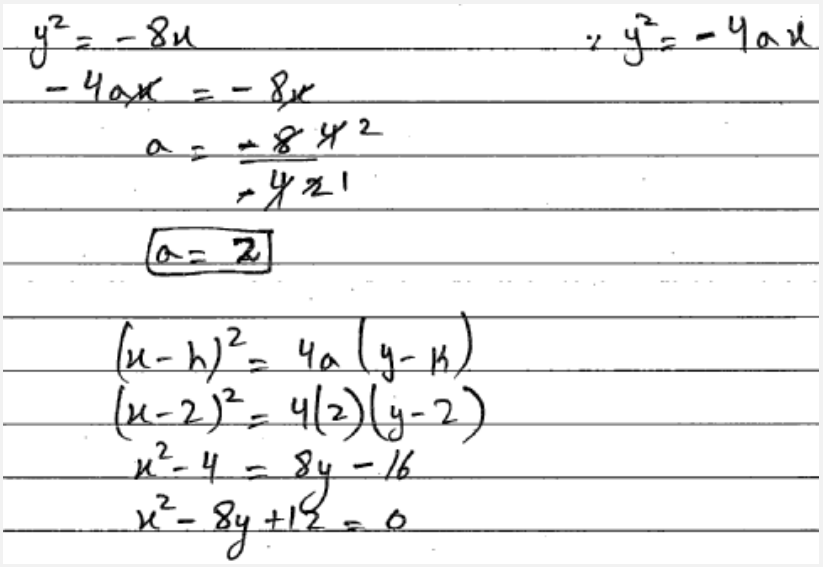
	solved this question partially correct. Candidates who had strong understanding of calculus and its application in analytical geometry were able to correctly find the equation of the normal to the parabola. However, other candidates struggled with applying the necessary concepts and calculations.
Description of Better Responses	Candidates who performed well demonstrated familiarity with the application of calculus in analytical geometry. They were able to use the correct method of finding the equations of tangent and normal to a curve (parabola) using the derivative. These candidates correctly differentiated the given equation of the parabola to find the slope of the tangent at the point (2,2), then used this information to find the slope of the normal. They accurately applied the point-slope form of a line equation to determine the equation of the normal and converted it to standard form without any errors in mathematical operations.
Image of Better Response	
Description of Weaker Responses	Weaker responses showed that many of the candidates were unfamiliar with the method of finding the equations of tangent and normal to a curve (parabola) using the derivative. They struggled to correctly differentiate the given equation or to apply the point-slope form to find the equation of the normal. This resulted in incorrect or incomplete solutions, showing that candidates need reinforcement of the application of calculus in analytical geometry.
Images of Weaker Responses	**Image (i)** 

Image (ii)

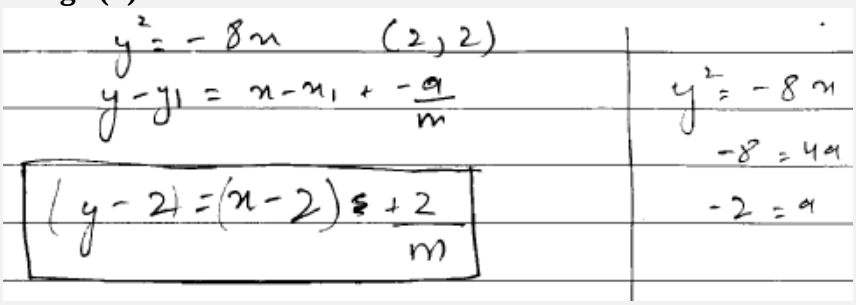
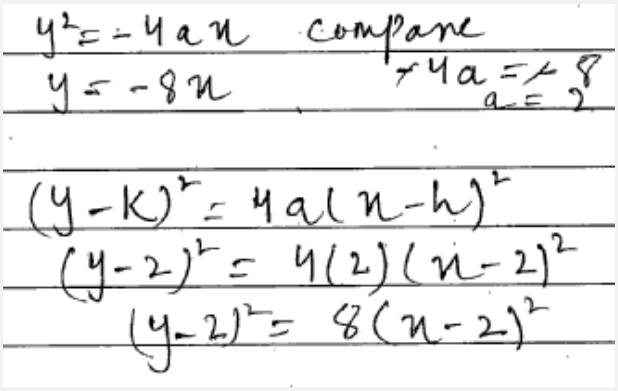


Image (iii)



Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

To enhance overall performance, consider the following strategies;

Practice Using Interactive Digital Tools:

- **GeoGebra:** Use GeoGebra to visualise the parabola and the normal line at a given point. This can help students to understand the geometric relationship between the curve, the tangent and the normal.
- **Desmos:** Utilise Desmos to plot the parabola and dynamically show the tangent and normal lines at various points. This interactive approach can help students grasp the concepts of differentiation and line equations.

Targeted Instruction:

- **Worked Examples:** Provide detailed examples that demonstrate the step-by-step process of finding the equation of the normal to a parabola using calculus. Highlight the differentiation process and the application of the point-slope form of a line equation.
- **Practice Problems:** Offer a variety of practice problems with different parabolas and points. Later, share the detailed solutions to reinforce the concepts and techniques.

- **Conceptual Understanding:** Emphasise the importance of understanding the geometric properties of parabolas and the role of derivatives in finding tangents and normals. Discuss the relationship between the slope of the tangent and the normal.

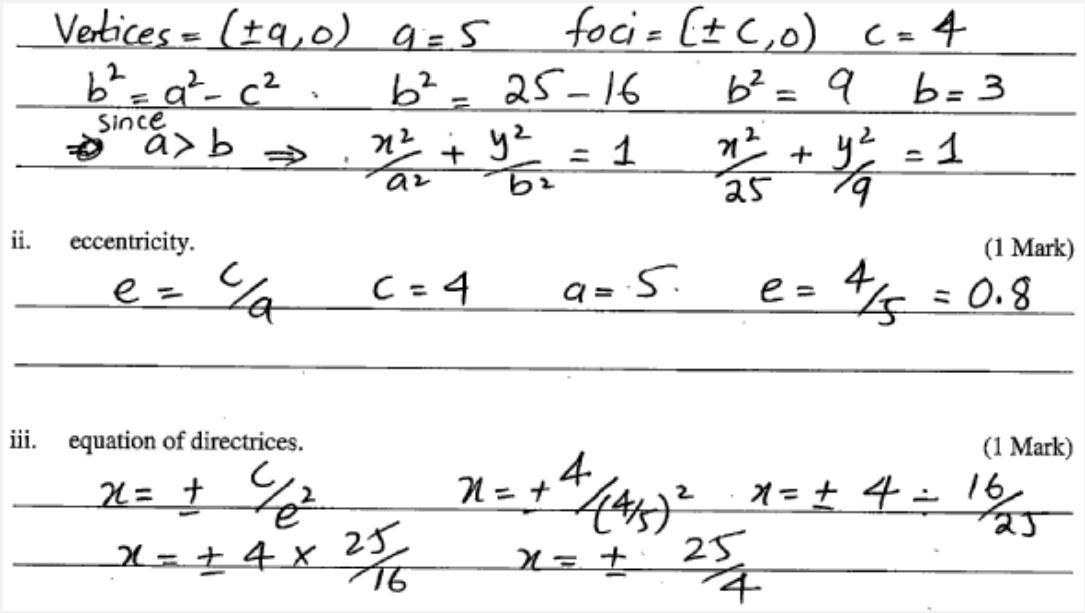
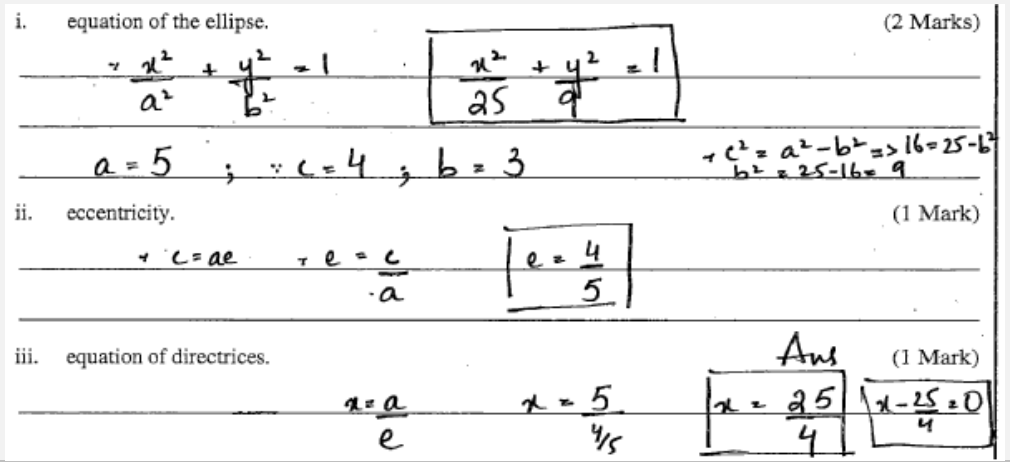
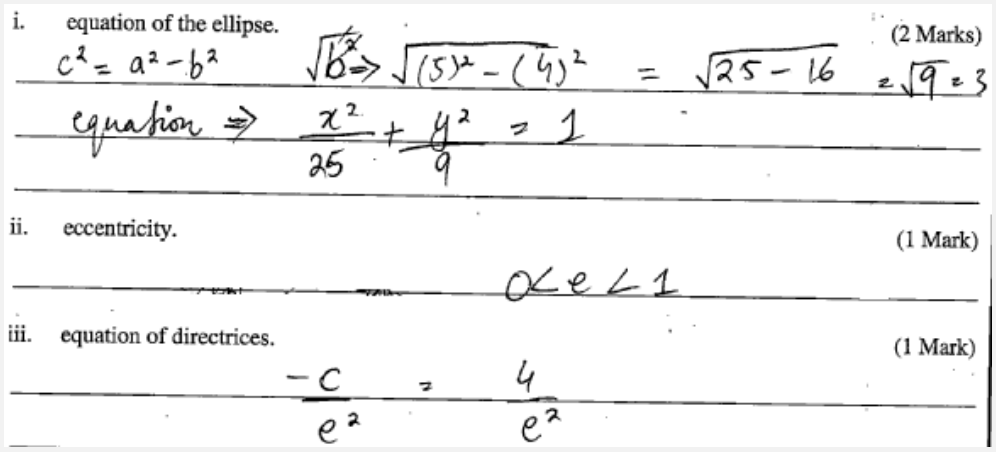
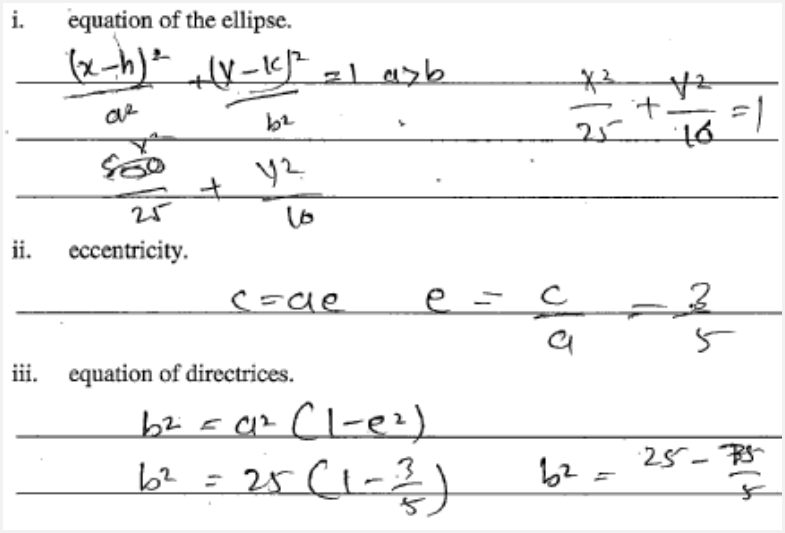
Question No. 7b **Candidates were given the choice to attempt any TWO out of the three questions: 7a, 7b, and 7c.**	
Question Text	An ellipse with centre $(0,0)$ has vertices $(\pm 5,0)$ and foci $(\pm 4,0)$. Find the i. equation of the ellipse. ii. eccentricity. iii. equation of directrices.
SLO No.	21.2.2/ 21.2.3
SLO Text	Find the equation of an ellipse with its given elements. Find the elements of ellipse by converting a given equation to the standard form.
Max Marks	4
Cognitive Level	A
Checking Hints	i. 1 mark for calculating the value of minor axis (*b*) 1 mark for substituting in the standard equation of ellipse ii. 1 mark for finding eccentricity iii. 1 mark for finding the equations of the directrices
Overall Performance	Overall, a good number of candidates attempted this question. Many of the candidates attempting this question scored full marks. The remaining candidates solved this question partially correct. Candidates who understood the terms and formulae related to ellipses were able to successfully determine the equation, eccentricity, and directrices. However, others struggled with recalling and applying these concepts accurately.
Description of Better Responses	Candidates who performed well demonstrated awareness of the terms and formulae of an ellipse. They correctly identified the standard form of the ellipse equation, using the given vertices to determine the lengths of the semi-major and semi-minor axes. They accurately calculated the eccentricity and wrote the correct equations for the directrices. These candidates showed a clear understanding of the relationship between the elements of an ellipse and applied mathematical operations without error.
Images of Better Responses	**Image (i)**  Vertices = $(\pm a, 0)$ $a = 5$ foci = $(\pm c, 0)$ $c = 4$ $b^2 = a^2 - c^2$ $b^2 = 25 - 16$ $b^2 = 9$ $b = 3$ since $a > b \Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $\frac{x^2}{25} + \frac{y^2}{9} = 1$ ii. eccentricity. (1 Mark) $e = c/a$ $c = 4$ $a = 5$ $e = 4/5 = 0.8$ iii. equation of directrices. (1 Mark) $x = \pm c/e^2$ $x = \pm 4/(4/5)^2$ $x = \pm 4 \div 16/25$ $x = \pm 4 \times 25/16$ $x = \pm 25/4$

	Image (ii)  i. equation of the ellipse. (2 Marks) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $\boxed{\frac{x^2}{25} + \frac{y^2}{9} = 1}$ $a = 5$; $c = 4$; $b = 3$ $c^2 = a^2 - b^2 \Rightarrow 16 = 25 - b^2$ $b^2 = 25 - 16 = 9$ ii. eccentricity. (1 Mark) $c = ae$ $e = \frac{c}{a}$ $\boxed{e = \frac{4}{5}}$ iii. equation of directrices. Ans (1 Mark) $x = \frac{a}{e}$ $x = \frac{5}{4/5}$ $\boxed{x = \frac{25}{4}}$ $\boxed{x - \frac{25}{4} = 0}$
Description of Weaker Responses	Weaker responses showed that some of the candidates lacked knowledge of terms and formulae related to ellipses. In part (i), some of the candidates struggled to identify the correct standard form of the ellipse equation, mistakenly applied the non-origin form, or made substitution errors leading to incorrect calculations. In part (ii), many were unable to accurately calculate the eccentricity, either by substituting incorrect values or by writing the condition for eccentricity instead of finding it. In part (iii), some of the candidates correctly wrote the equation of the directrices however, they were unable to substitute the values, while others wrote incorrect equations for the directrices. These issues highlight the need for reinforcement in solving problems related to geometric properties and algebraic methods.
Images of Weaker Responses	**Image (i)**  i. equation of the ellipse. (2 Marks) $c^2 = a^2 - b^2$ $\sqrt{b^2} \Rightarrow \sqrt{(5)^2 - (4)^2} = \sqrt{25 - 16} = \sqrt{9} = 3$ equation $\Rightarrow \frac{x^2}{25} + \frac{y^2}{9} = 1$ ii. eccentricity. (1 Mark) $0 < e < 1$ iii. equation of directrices. (1 Mark) $\frac{-c}{e^2} = \frac{4}{e^2}$ **Image (ii)**  i. equation of the ellipse. $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$ $a > b$ $\frac{x^2}{25} + \frac{y^2}{16} = 1$ $\frac{y^2}{25} + \frac{y^2}{16}$ ii. eccentricity. $c = ae$ $e = \frac{c}{a} = \frac{3}{5}$ iii. equation of directrices. $b^2 = a^2(1 - e^2)$ $b^2 = 25\left(1 - \frac{3}{5}\right)$ $b^2 = 25 - \frac{75}{5}$

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

To enhance overall performance, consider the following strategies;

Practice Using Interactive Digital Tools:

- **GeoGebra:** Use GeoGebra to visualise ellipses and their elements. This can help students to understand the geometric relationships between the centre, vertices, foci and directrices.
- **Desmos:** Utilise Desmos to plot ellipses and dynamically show the effect of changing the vertices and foci on the shape and elements of the ellipse. This interactive approach can help students grasp the concepts of ellipse equations and properties.

Targeted Instruction:

- **Worked Examples:** Provide detailed examples that demonstrate the step-by-step process of finding the equation, eccentricity and directrices of an ellipse. Highlight the use of given elements to determine the necessary lengths and formulae.
- **Practice Problems:** Offer a variety of practice problems with different configurations of ellipses and their elements. Include detailed solutions to reinforce the concepts and techniques.
- **Conceptual Understanding:** Emphasise the importance of understanding the geometric properties of ellipses and the algebraic methods used to find their elements. Discuss the relationship between the lengths of the semi-major and semi-minor axes, eccentricity and directrices.

Question No. 7c **Candidates were given the choice to attempt any TWO out of the three questions: 7a, 7b, and 7c.**	
Question Text	The vertices of a hyperbola are $(15, 1)$ and $(-1, 1)$. If the length of its conjugate axis is 12 units, then find the equation of the hyperbola in standard form.
SLO No.	22.3.2
SLO Text	Find the equation of tangent and normal to a hyperbola (a) at a point.
Max Marks	4
Cognitive Level	U
Checking Hints	1 mark for finding h 1 mark for finding a 1 mark for finding b 1 mark for finding equation

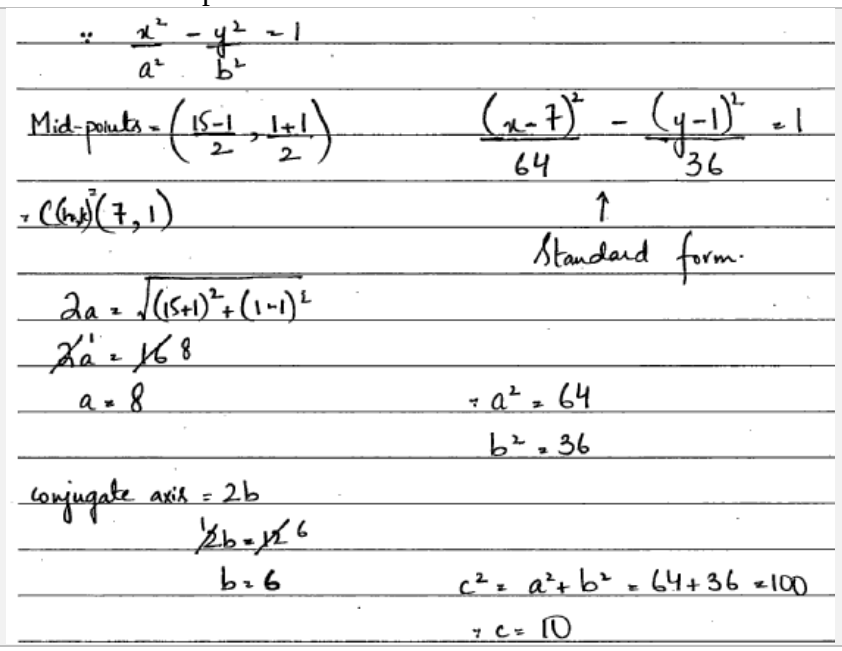
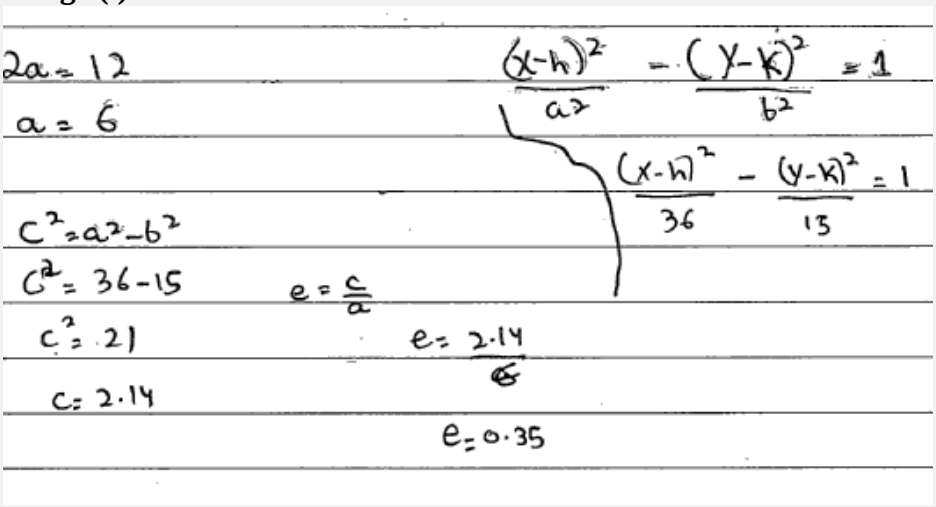
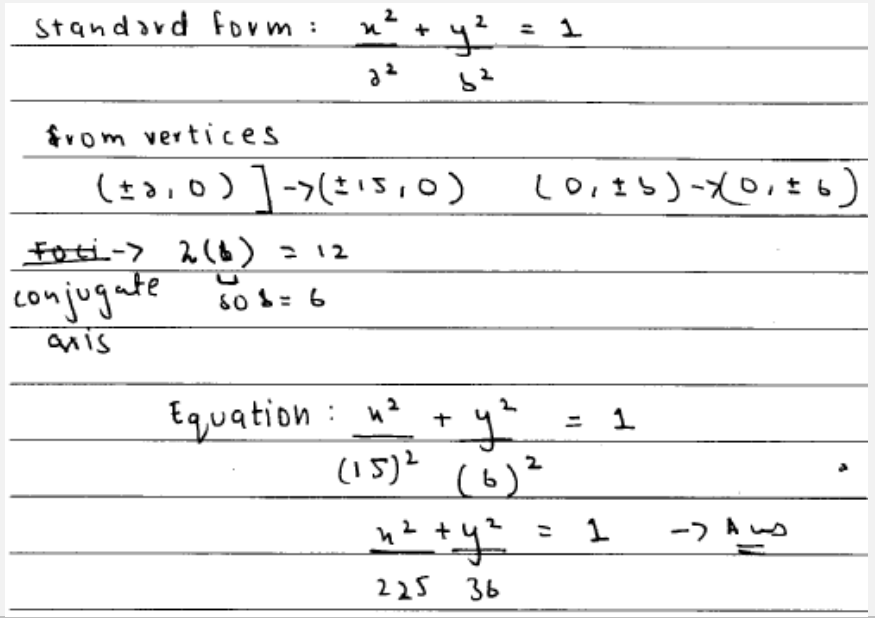
Overall Performance	Overall, a good number of candidates attempted this question and many of the candidates attempting this question scored full marks. The remaining candidates solved this question partially correct. Candidates who understood the terms and formulae related to hyperbolas were able to accurately determine the equation in standard form. However, others struggled with recalling and applying these concepts correctly.
Description of Better Responses	Candidates who performed well demonstrated a strong understanding of the terms and formulae of a hyperbola. These candidates correctly identified the standard form of the hyperbola equation. They applied the midpoint and distance formula to the given vertices to determine the center and the length of the semi-transverse axis. They accurately calculated the necessary values, including the lengths of the semi-conjugate axis, and used these to write the equation of the hyperbola in standard form. These candidates demonstrated a clear understanding of the required geometric properties and algebraic methods, performing mathematical operations without error.
Image of Better Response	
Description of Weaker Responses	In weaker responses, some of the candidates struggled to identify the correct semi-conjugate axes, mistakenly taking it as 2*a* instead of 2*b*, leading to incorrect values for *a*. Also many of the candidates were not able to apply the midpoint and the distance of the given vertices to find the center and the length of the semi-transverse axes or the major axes. Additionally, a few candidates incorrectly substituted the values of *a* and *c* as 6and 15 to find the value of *c* and calculated the eccentricity, which was unnecessary. Some even wrote the standard form of the ellipse instead of the hyperbola. These errors resulted in incorrect or incomplete equations, indicating a lack of understanding of the terms and formulae related to hyperbolas.
Images of Weaker Responses	**Image (i)** 

Image (ii)



Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

To enhance overall performance, consider the following strategies;

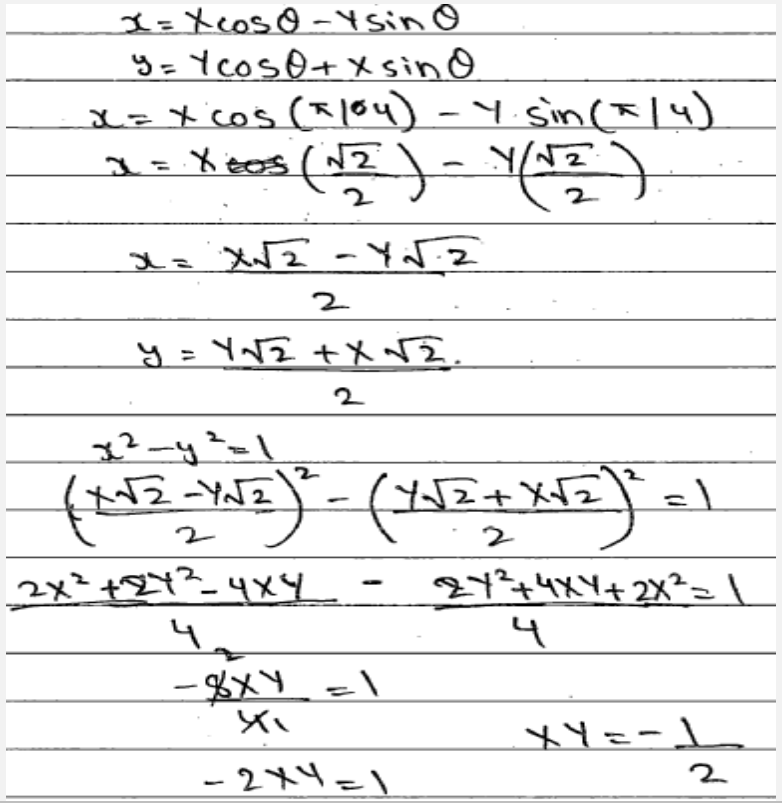
Practice Using Interactive Digital Tools:

- **GeoGebra:** Use GeoGebra to visualise hyperbolas and their elements. This can help students to understand the geometric relationships between the vertices, foci and axes of the hyperbola.
- **Desmos:** Utilise Desmos to plot hyperbolas and dynamically show the effect of changing the vertices and conjugate axis on the shape and elements of the hyperbola. This interactive approach can help students grasp the concepts of hyperbola equations and properties.

Targeted Instruction:

- **Worked Examples:** Provide detailed examples that demonstrate the step-by-step process of finding the equation of a hyperbola in standard form. Highlight the use of given elements to determine the necessary lengths and formulae.
- **Practice Problems:** Offer a variety of practice problems with different configurations of hyperbolas and their elements. Include detailed solutions to reinforce the concepts and techniques.

- **Conceptual Understanding:** Emphasise the importance of understanding the geometric properties of hyperbolas and the algebraic methods to find their elements. Discuss the relationship between the lengths of the semi-transverse and semi-conjugate axes and their use in forming the equation of the hyperbola.

Question No. 8	
Question Text	Find the equation of the hyperbola $x^2 - y^2 = 1$ with respect to new axes obtained by rotation of axes about the origin through an angle of $\frac{\pi}{4}$ radians.
SLO No.	23.1.3
SLO Text	Find the transformed equation by using translation or rotation of axes.
Max Marks	4
Cognitive Level	A
Checking Hints	1 mark for each equation of transformation for *x* 1 mark for each equation of transformation for *y* 1 mark for substitution of equations of transformation into given equation 1 mark for new equation
Overall Performance	Overall, candidates struggled in solving this question. Candidates who understood the concepts and formulae related to the rotation of axes were able to successfully derive the transformed equation. However, others struggled with applying these concepts accurately.
Description of Better Responses	Candidates who performed well were attentive to the equations of rotational axes. They correctly identified the transformation equations for *x* and *y* and used the given angle π/4 to find the new coordinates for *x* and *y*. These candidates then substituted the new coordinates into the original equation to obtain the transformed equation. They demonstrated a clear understanding of applying the rotation and solving the resultant simultaneous equations accurately, without errors in mathematical operations.
Image of Better Response	

$x = X\cos\theta - Y\sin\theta$

$y = Y\cos\theta + X\sin\theta$

$x = X\cos(\pi/4) - Y\sin(\pi/4)$

$x = X\left(\frac{\sqrt{2}}{2}\right) - Y\left(\frac{\sqrt{2}}{2}\right)$

$x = \frac{X\sqrt{2} - Y\sqrt{2}}{2}$

$y = \frac{Y\sqrt{2} + X\sqrt{2}}{2}$

$x^2 - y^2 = 1$

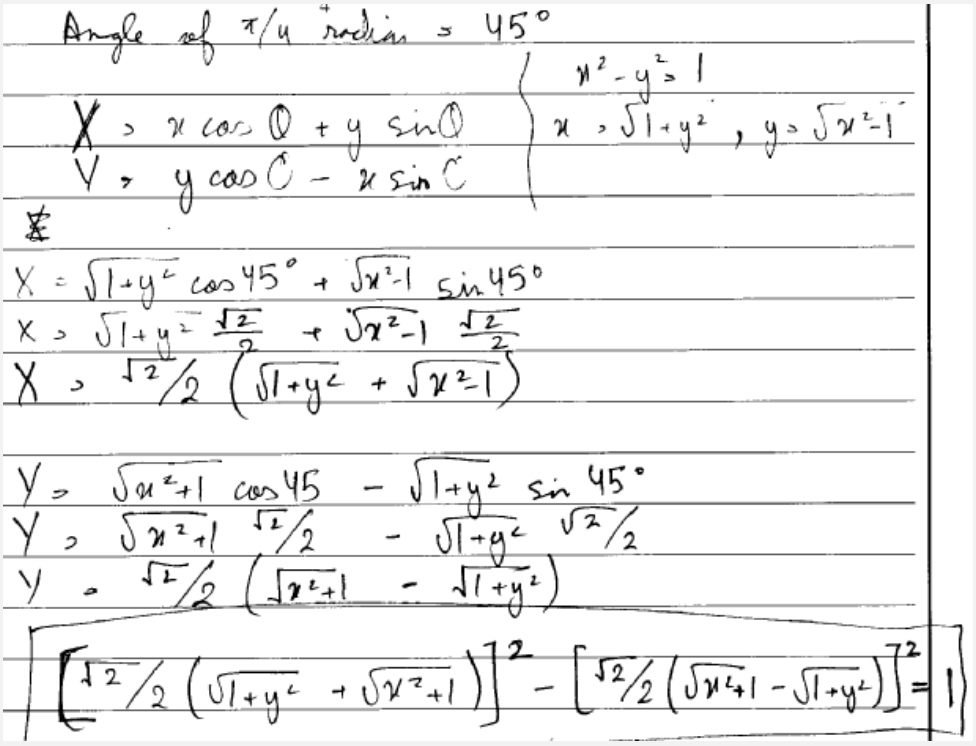
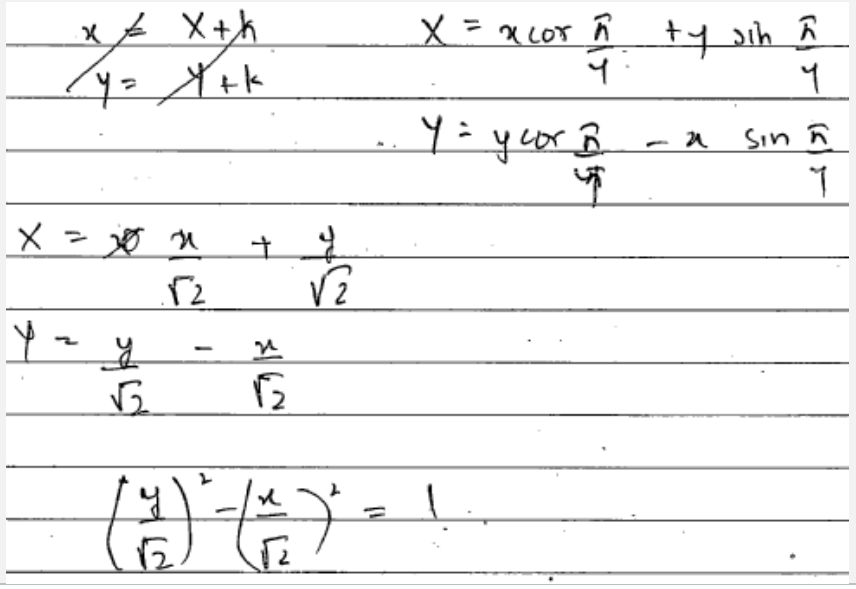
$\left(\frac{X\sqrt{2} - Y\sqrt{2}}{2}\right)^2 - \left(\frac{Y\sqrt{2} + X\sqrt{2}}{2}\right)^2 = 1$

$\frac{2X^2 + 2Y^2 - 4XY}{4} - \frac{2Y^2 + 4XY + 2X^2}{4} = 1$

$\frac{-8XY}{4} = 1$

$-2XY = 1$

$XY = -\frac{1}{2}$

Description of Weaker Responses	In weaker response many candidates struggled with the equations for rotational axes. They used $X = x\cos\theta + y\sin\theta$, and $Y = y\cos\theta - x\sin\theta$ instead of $X = x\cos\theta - y\sin\theta$ and $Y = y\cos\theta + x\sin\theta$. They substituted the given angle to calculate the new coordinates but incorrectly substituted x as $\frac{y^2}{\sqrt{2}}$ and y as $\frac{x^2}{\sqrt{2}}$, without solving further. Additionally, some of the candidates substituted x and y as $\sqrt{1+y^2}$ and $\sqrt{1+x^2}$ in the new coordinates and then substituted these into the given equation to find the new equation. This indicates a lack of understanding of the rotation and the steps needed to transform the equation, leading to incorrect or incomplete solutions.
Images of Weaker Responses	**Image (i)**  Angle of $\pi/4$ radian $= 45°$ $X = x\cos\theta + y\sin\theta$ $Y = y\cos\theta - x\sin\theta$ $x^2 - y^2 = 1$ $x = \sqrt{1+y^2}$, $y = \sqrt{x^2-1}$ $X = \sqrt{1+y^2}\cos 45° + \sqrt{x^2-1}\sin 45°$ $X = \sqrt{1+y^2}\frac{\sqrt{2}}{2} + \sqrt{x^2-1}\frac{\sqrt{2}}{2}$ $X = \sqrt{2}/2\left(\sqrt{1+y^2} + \sqrt{x^2-1}\right)$ $Y = \sqrt{x^2+1}\cos 45 - \sqrt{1+y^2}\sin 45°$ $Y = \sqrt{x^2+1}\sqrt{2}/2 - \sqrt{1+y^2}\sqrt{2}/2$ $Y = \sqrt{2}/2\left(\sqrt{x^2+1} - \sqrt{1+y^2}\right)$ $\left[\sqrt{2}/2\left(\sqrt{1+y^2} + \sqrt{x^2+1}\right)\right]^2 - \left[\sqrt{2}/2\left(\sqrt{x^2+1} - \sqrt{1+y^2}\right)\right]^2 = 1$ **Image (ii)**  $x = X + h$ $y = Y + k$ $X = x\cos\frac{\pi}{4} + y\sin\frac{\pi}{4}$ $Y = y\cos\frac{\pi}{4} - x\sin\frac{\pi}{4}$ $X = \frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}}$ $Y = \frac{y}{\sqrt{2}} - \frac{x}{\sqrt{2}}$ $\left(\frac{y}{\sqrt{2}}\right)^2 - \left(\frac{x}{\sqrt{2}}\right)^2 = 1$

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

To enhance overall performance, consider the following strategies;

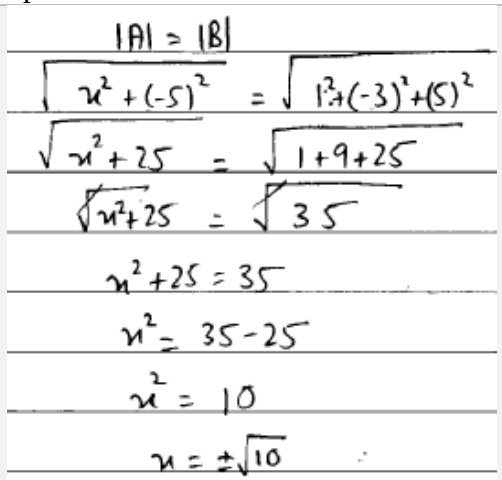
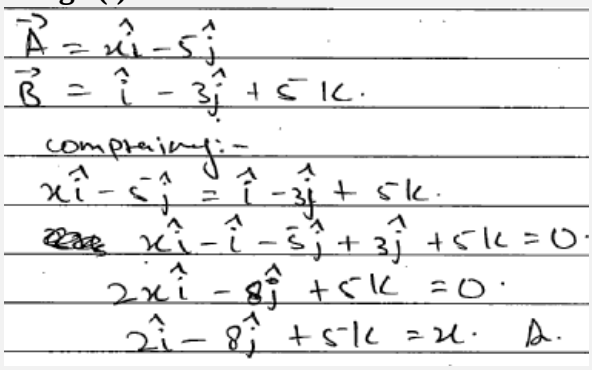
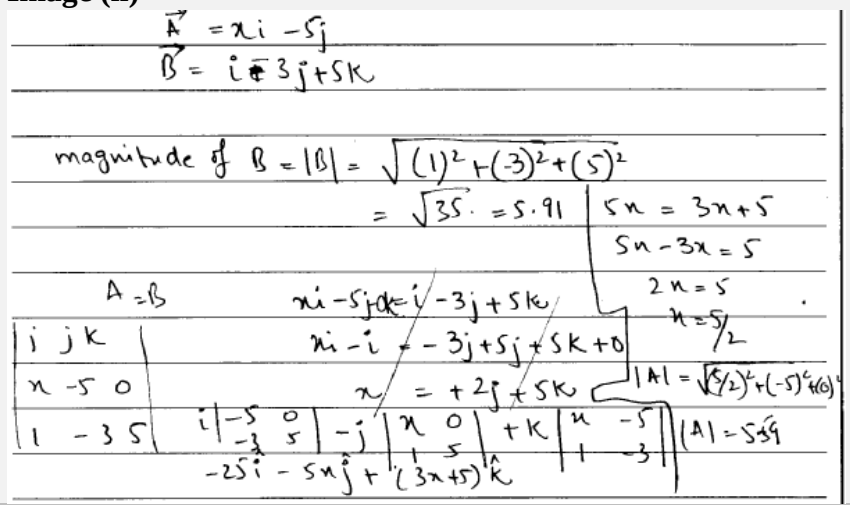
Practice Using Interactive Digital Tools:

- **GeoGebra:** Use GeoGebra to visualise the rotation of axes and its effects on the equation of a conic section. This can help students to understand the geometric impact of rotation on the equation.
- **Desmos:** Utilise Desmos to dynamically demonstrate the rotation of hyperbolas and other conic sections about the origin. This interactive approach can help students grasp the concepts of coordinate transformation.

Targeted Instruction:

- **Worked Examples:** Provide detailed examples that demonstrate the step-by-step process of finding the transformed equation through rotation of axes. Highlight the use of the rotation matrix and the substitution of new coordinates.
- **Practice Problems:** Offer a variety of practice problems involving different angles of rotation and types of conic sections. Include detailed solutions to reinforce the concepts and techniques.
- **Conceptual Understanding:** Emphasise the importance of understanding the geometric properties of conic sections and the algebraic methods used for rotation of axes. Discuss the derivation and application of the rotation matrix in transforming equations.

Question No. 9a **Candidates were given the choice to attempt any ONE out of the two questions: 9a and 9b.**	
Question Text	Find the value of x for which the two vectors $\vec{A} = x\hat{i} - 5\hat{j}$ and $\vec{B} = \hat{i} - 3\hat{j} + 5\hat{k}$ have same magnitudes.
SLO No.	24.2.5
SLO Text	Find the magnitude of a vector.
Max Marks	3
Cognitive Level	U
Checking Hints	1 mark for equalising the magnitudes 1 mark for substituting the correct values in magnitude on both sides 1 mark for finding the value of x
Overall Performance	Most candidates attempted this question successfully. Candidates who had a clear understanding of vector magnitudes and the process of equating them, were able to find the

	correct value of *x*. However, others struggled with these concepts and made errors in their calculations.
Description of Better Responses	Candidates who performed well demonstrated a strong understanding of vector magnitudes. They correctly calculated the magnitudes of the given vectors *A* and *B* and set them equal to find the value of *x*. These candidates showed proficiency in handling the mathematical operations and arrived at the correct solution without any errors.
Image of Better Response	
Description of Weaker Responses	Many candidates struggled to calculate the correct magnitudes of the given vectors and were unable to find the value of *x*. Some compared the vectors directly without calculating their magnitudes, resulting in incorrect equations for *x*. Others correctly calculated the magnitude of vector *B* however, left the solution incomplete. Additionally, a few candidates applied the cross product to vectors *AB* and did not complete the expansion. These issues indicate a gap in understanding basic vector properties and operations, leading to incorrect or incomplete solutions.
Images of Weaker Responses	**Image (i)**  **Image (ii)** 

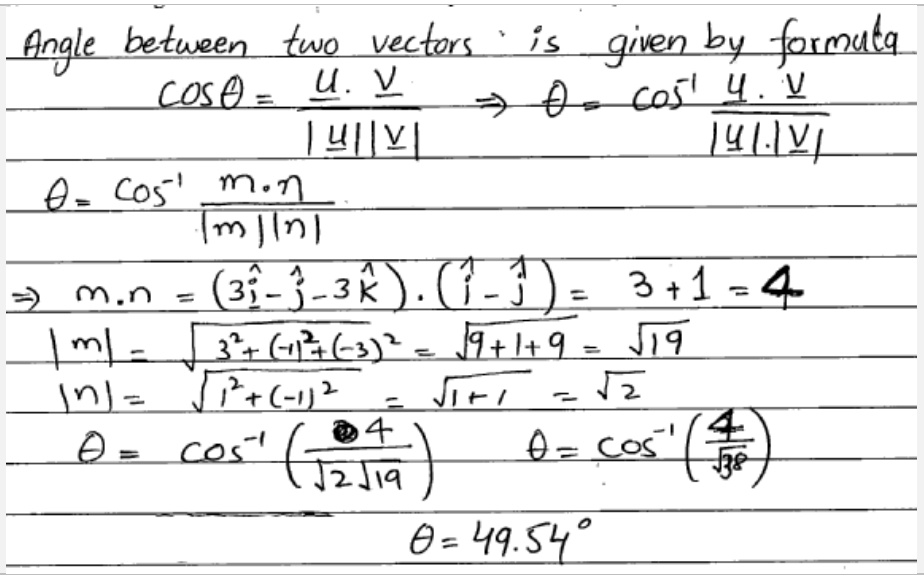
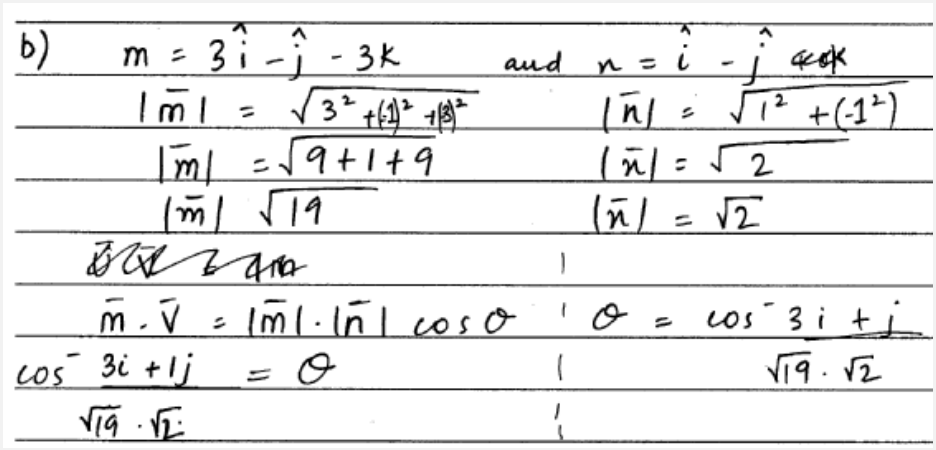
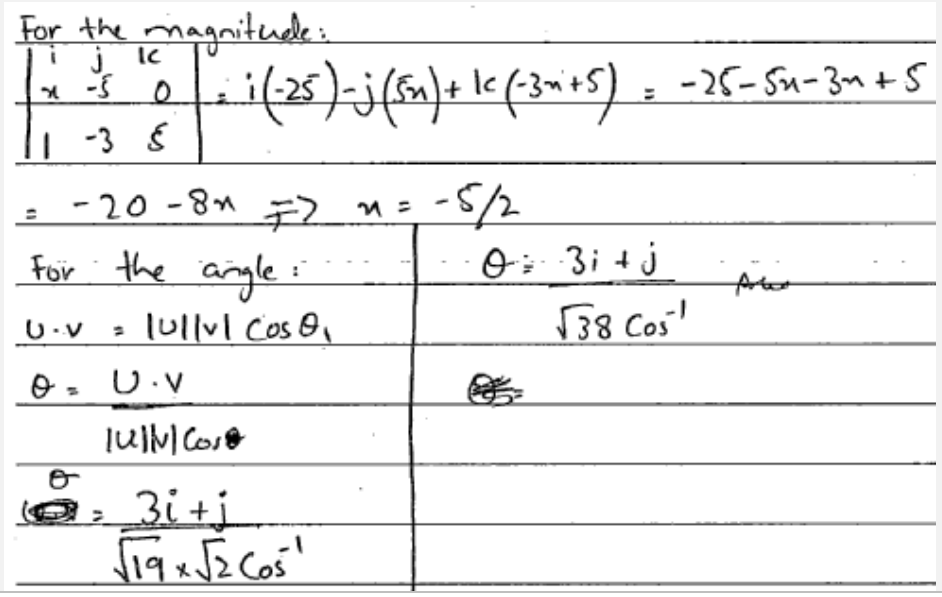
Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:
To enhance overall performance, consider the following strategies;
Practice Using Interactive Digital Tools:

- **GeoGebra:** Use GeoGebra to visualise vectors and their magnitudes. This can help students understand the concept of vector magnitude and the geometric representation of vectors.
- **Desmos:** Utilise Desmos to plot vectors and dynamically demonstrate the effect of changing components on the magnitude of vector. This interactive approach can help students grasp the relationship between vector components and magnitude

Question No. 9b **Candidates were given the choice to attempt any ONE out of the two questions: 9a and 9b.**	
Question Text	Find the angle between the vectors $m = 3\hat{i} - \hat{j} - 3\hat{k}$ and $n = \hat{i} - \hat{j}$.
SLO No.	24.3.9
SLO Text	Apply dot product to find the angle between two vectors.
Max Marks	3
Cognitive Level	A
Checking Hints	1 mark for finding the magnitudes 1 mark for finding the dot product on both sides 1 mark for finding the angle
Overall Performance	Most of the candidates primarily attempted this question. The overall performance of candidates on this question shows considerable variability. Candidates who understood the concept of the dot product and its application, were able to find the angle between two vectors and solve given the problem correctly. However, others struggled with these concepts and made errors in their calculations.
Description of Better Responses	Candidates who performed well had a clear understanding of the formula for the angle between two vectors. They were able to calculate the magnitudes of the vectors and the dot product between them. These candidates correctly applied the formula $\cos\theta = \frac{m.n}{\lvert m\rvert\lvert n\rvert}$ to find the angle θ. They performed the mathematical operations without any errors and obtained the correct angle.

Image of Better Response	
Description of Weaker Responses	Many candidates applied the correct formula and calculated the magnitudes correctly; however, they either did not find the dot product or used the cross product instead. This led to incomplete or incorrect calculations of the angle between the two vectors. This indicates a need for thorough reinforcement of vector properties and the correct application of the dot product.
Images of Weaker Responses	**Image (i)**  **Image (ii)** 

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login

Any Additional Suggestion:

To enhance overall performance, consider the following strategies:

Practice Using Interactive Digital Tools:

- **GeoGebra:** Use GeoGebra to visualise vectors and their dot products. This can help students to understand the geometric interpretation of the dot product and its relation with the angle between the vectors.
- **Desmos:** Utilise Desmos to plot vectors and dynamically show the variation in the angle between them as the dot product changes. This interactive approach can help students grasp the concept of vector angles and the role of the dot product.

Annexure A: Pedagogies Used for Teaching the SLOs

Pedagogy: Storyboard
Description: A visual pedagogy that uses a series of illustrated panels to present a narrative, encouraging creativity and critical thinking. It helps learners organise ideas, sequence events, and comprehend complex concepts through storytelling.
Example: In a Literature class, students are tasked with creating storyboards to visually retell a novel. They draw key scenes, write captions, and present their stories to the class, enhancing their reading comprehension and fostering their imagination.

Pedagogy: Cause and Effect
Description: This pedagogy explores the relationships between actions and consequences. By analysing cause-and-effect relationships, learners develop a deeper understanding of how events are interconnected and how one action can lead to various outcomes.
Example: In a History class, students study the causes and effects of the Industrial Revolution. They research and discuss how technological advancements in manufacturing led to significant societal changes, such as urbanisation and labour reform movements.

Pedagogy: Fish and Bone
Description: A method that breaks down complex topics into main ideas (the fish) and supporting details (the bones). This visual approach enhances comprehension by highlighting essential concepts and their relevant explanations.
Example: During a Biology class on human anatomy, the teacher uses the fish and bone technique to teach about the human skeletal system. Teacher presents the main components of the human skeleton (fish) and elaborates on each bone's structure and function (bones).

Pedagogy: Concept Mapping
Description: An effective way to visually represent relationships between ideas. Learners create diagrams connecting key concepts, aiding in understanding the overall structure of a subject and fostering retention.
Example: In a Psychology assignment, students use concept mapping to explore the various theories of personality. They interlink different theories, such as Freud's psychoanalysis, Jung's analytical psychology, and Bandura's social-cognitive theory, to see how they relate to each other.

Pedagogy: Audio Visual Resources
Description: Incorporating multimedia elements like videos, images, and audio into lessons. This approach caters to different learning styles, making educational content more engaging and memorable.
Example: In a General Science class, the teacher uses a documentary-style video to teach about the solar system. The video includes stunning visual animations of the planets, interviews with astronomers, and background music, enhancing students' interest and understanding of space.

Pedagogy: Think, Pair, and Share
Description: A collaborative learning technique where students ponder a question or problem individually, then discuss their thoughts in pairs or small groups before sharing with the entire class. It fosters active participation, communication skills, and diverse perspectives.
Example: In a Literature in English class, the teacher poses a thought-provoking question about a novel's moral dilemma. Students first reflect individually, then pair up to exchange their opinions, and finally participate in a lively class discussion to explore different viewpoints.

Pedagogy: Questioning Technique (Socratic Approach)
Description: Based on Socratic dialogue, this method stimulates critical thinking by posing thought-provoking questions. It encourages learners to explore ideas, justify their reasoning, and discover knowledge through a process of inquiry.
Example: In an Ethics class, the instructor uses the Socratic approach to lead a discussion on the meaning of justice. By asking a series of probing questions, the students engage in a deeper exploration of ethical principles and societal values.

Pedagogy: Practical Demonstration
Description: A hands-on approach where learners observe real-life applications of theories or skills. Practical demonstrations enhance comprehension, skill acquisition, and problem-solving abilities by bridging theoretical concepts with real-world scenarios.
Example: In a Food and Nutrition class, the instructor demonstrates the proper technique for filleting a fish. Students observe and then practice the skill themselves, learning the practical application of knife skills and culinary precision.

(**Note:** The examples provided in this annexure serve as illustrations of various pedagogies. It is important to understand that these pedagogies are versatile and can be applied across subjects in numerous ways. Feel free to adapt and explore these techniques creatively to enhance learning outcomes in your specific context.)

Acknowledgements

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Additionally, we express our gratitude to the esteemed team of reviewers for their constructive feedback on overall performance, better and weaker responses, and validating teaching pedagogies along with suggestions for improvement.
These contributors include:

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- Dr Naveed Yousuf, CEO, AKU-EB